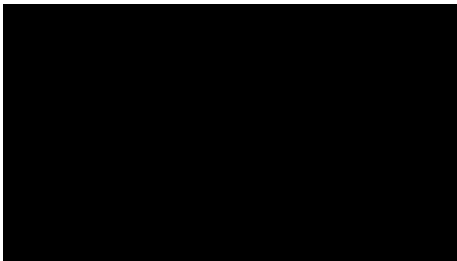


Geo1: Origins of the Universe

*Each lesson in this course begins with independent study using the course eBook. These eBooks are both the 'lectures' and textbook for this course. They introduce the basic concepts for each topic that we will study, and provide links to more in depth videos and articles. **You are required to view the videos and read these articles to complete your study.** To access the eBook click on the link below. The Table of Contents and the Index are provided to help you access the content quickly in the future (for review or reference purposes)*



In order to understand how our planet formed, changed, and currently works, we need to learn about the universe as a whole. Where did the universe come from? This eBook introduces some of the theories that try to explain the origins of the universe.

In this week's lab, we will introduce 'Earth Systems Science' and explore some of the basic properties of the universe.

Table of Contents

- [1 Origins of the Universe \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-1-origins-of-the-universe)
- [2 The Big Bang Theory \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check)
- [3 Big Bang Theory Self Check \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check)
- [4 Evidence Supporting the Big Bang Theory \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-4-evidence-supporting-the-big-bang-theory\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-4-evidence-supporting-the-big-bang-theory)
- [5 What is Energy? \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-5-what-is-energy\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-5-what-is-energy)
- [6-7 The Range of Energy: The Electromagnetic Spectrum \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-6-the-range-of-energy-the-electromagnetic-spectrum\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-6-the-range-of-energy-the-electromagnetic-spectrum)
- [8 Doppler Shift and Hubble's Law \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-8-doppler-shift-and-hubble's-law\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-8-doppler-shift-and-hubble's-law)

[8-doppler-shift-and-hubbles-law\)](#)

9 Hubble's Law and the Slope of a Straight Line

[\(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-9-hubbles-law-and-the-slope-of-a-straight-line\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-9-hubbles-law-and-the-slope-of-a-straight-line)

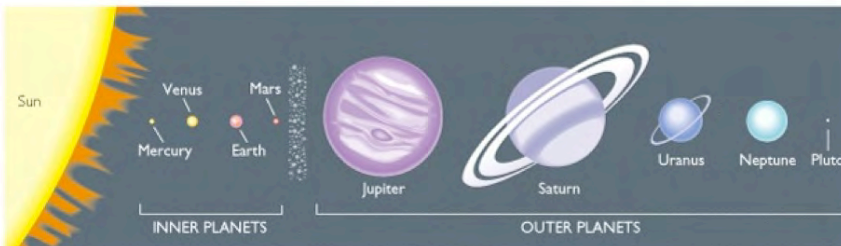
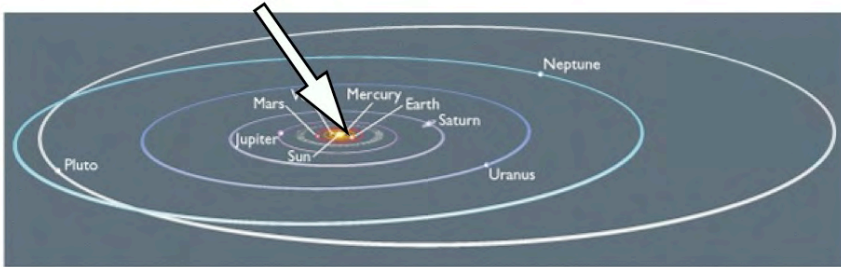
Index of Readings & Videos

- [M. Turner. "Origin of the Universe," *Scientific American* \(2013\)](#)
[\(https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?wrap=1\)](https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?wrap=1)
- [NASA Astrophysics: The Big Bang](http://science.nasa.gov/astrophysics/focus-areas/what-powered-the-big-bang/) (<http://science.nasa.gov/astrophysics/focus-areas/what-powered-the-big-bang/>)
- [NASA's page on the Electromagnetic Spectrum](http://imagine.gsfc.nasa.gov/science/toolbox/emspectrum1.html) ⇨ (<http://imagine.gsfc.nasa.gov/science/toolbox/emspectrum1.html>)
- [NASA's page on Hubble's Law](http://imagine.gsfc.nasa.gov/features/yba/M31-velocity/hubble-more.html) ⇨ (<http://imagine.gsfc.nasa.gov/features/yba/M31-velocity/hubble-more.html>)

Start eBook -> (<https://catcourses.ucmerced.edu/courses/32116/pages/our-planetary-neighborhood>)

Our Planetary Neighborhood

You Are Here



In our solar system, the planets make up only about 0.15% of our solar system mass. The sun accounts for the other 99.85% mass! The four inner planets, Mercury, Venus, Earth, and Mars, are the terrestrial or "rocky" planets, with dense, solid cores, while the outer planets are the Jovian or "gas giants" made mostly of the light elements hydrogen and helium.

How did our solar system, and the Universe, come into existence?

1

Geo1 TOC (<https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe>).

Next -> (<https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-2-big-bang-theory>).



COSMOLOGY

ORIGIN OF THE UNIVERSE

Cosmologists are closing in on the ultimate processes that created and shaped the universe

By Michael S. Turner

THE UNIVERSE IS BIG IN BOTH SPACE AND TIME AND, FOR MUCH OF humankind's history, was beyond the reach of our instruments and our minds. That changed dramatically in the 20th century. The advances were driven equally by powerful ideas—from Einstein's general relativity to modern theories of the elementary particles—and powerful instruments—from the 100- and 200-inch reflectors that George Ellery Hale built, which took us beyond our Milky Way galaxy, to the Hubble Space Telescope, which has taken us back to the birth of galaxies. Over the past 30 years the pace of progress has accelerated with the realization that dark matter is not made of ordinary atoms, the discovery of dark energy, and the dawning of bold ideas such as cosmic inflation and the multiverse.

IN BRIEF

Our universe began with a hot big bang 13.7 billion years ago and has expanded and cooled ever since. It has evolved from a formless soup of elementary particles into the richly structured cosmos of today.

The first microsecond was the formative period when matter came to dominate over antimatter, the seeds for galaxies and other structures were planted, and dark matter (the unidentified material

that holds those structures together) was created. **The universe's future** lies in the hands of dark energy, an unknown form of energy that caused cosmic expansion to start accelerating a few billion years ago.

GENE BURKHARDT (styling)

E
X
T
R
E
M
E

P
H
Y
S
I
C
S



The universe of 100 years ago was simple: eternal, unchanging, consisting of a single galaxy, containing a few million visible stars. The picture today is more complete and much richer. The cosmos began 13.7 billion years ago with the big bang. A fraction of a second after the beginning, the universe was a hot, formless soup of the most elementary particles, quarks and leptons. As it expanded and cooled, layer on layer of structure developed: neutrons and protons, atomic nuclei, atoms, stars, galaxies, clusters of galaxies, and finally superclusters. The observable part of the universe is now inhabited by 100 billion galaxies, each containing 100 billion stars and probably a similar number of planets. Galaxies themselves are held

together by the gravity of the mysterious dark matter. The universe continues to expand and indeed does so at an accelerating pace, driven by dark energy, an even more mysterious form of energy whose gravitational force repels rather than attracts.

The overarching theme in our universe's story is the evolution from the simplicity of the quark soup to the complexity we see today in galaxies, stars, planets and life. These features emerged one by one over billions of years, guided by the basic laws of physics. In our journey back to the beginning of creation, cosmologists first travel through the well-established history of the universe back to the first microsecond; then to within 10^{-34} second of the beginning,

for which ideas are well formed but the evidence is not yet firm; and finally to the earliest moments of creation, for which our ideas are still just speculation. Although the ultimate origin of the universe still lies beyond our grasp, we have tantalizing conjectures, including the notion of the multiverse, whereby the universe comprises an infinite number of disconnected subuniverses.

EXPANDING UNIVERSE

USING THE 100-INCH HOOKER TELESCOPE ON Mount Wilson in 1924, Edwin Hubble showed that fuzzy nebulae, studied and speculated about for several hundred years, were galaxies just like our own—thereby enlarging the known universe by 100 billion. A few years later he showed that galaxies are moving apart from one another in a regular pattern described by a mathematical relation now known as Hubble's law, according to which galaxies that are farther away are moving faster. It is Hubble's law, played back in time, that points to a big bang 13.7 billion years ago.

Hubble's law found ready interpretation within general relativity: space itself is expanding, and galaxies are being carried along for the ride [see box on opposite page]. Light, too, is being stretched, or redshifted—a process that saps its energy, so that the universe cools as it expands. Cosmic expansion provides the narrative for understanding how today's universe came to be. As cosmologists imagine rewinding the clock, the universe becomes denser, hotter, more extreme and simpler. In exploring the beginning, we also probe the inner workings of nature by taking advantage of an accelerator more powerful than any built on Earth—the big bang itself.

By looking out into space with telescopes, astronomers peer back in time—and the larger the telescope, the farther back they peer. The light from distant galaxies reveals an earlier epoch, and the amount this light has redshifted indicates how much the universe has grown in the intervening years. The current record holder has a redshift of more than 10, representing a time when the universe was less than one-eleventh its present size and only a few hundred million years old. Telescopes such as the Hubble Space Telescope and the 10-meter Keck telescopes on Mauna Kea rou-

COSMIC TIMELINE

Before the Big Bang

Cosmologists do not yet know how the universe began, but this question has now come within the realm of science, with a number of speculative scenarios being discussed.

No Previous Era

Matter, energy, space and time begin abruptly with the big bang

Quantum Emergence

Ordinary space and time develop out of a primeval state described by a quantum theory of gravity

Multiverse

Our universe and others bud off from eternal space

Cyclic Universe

The big bang is the latest stage in an eternal cycle of expansion, collapse and renewed expansion

10^{-43} Second

Planck era: earliest meaningful time; space and time take shape

(From previous cycle)

E
X
T
R
E
M
E



P
H
Y
S
I
C
S

tinely take us back to the epoch when galaxies like ours were forming, a few billion years after the big bang. Light from even earlier times is so strongly redshifted that astronomers must look for it in the infrared and radio bands. Telescopes such as the planned James Webb Space Telescope, a 6.5-meter infrared telescope, and the Atacama Large Millimeter Array (ALMA), a network of 66 radio dishes already operating in northern Chile, can take us back to the birth of the very first stars and galaxies.

Computer simulations say that those stars and galaxies emerged when the universe was about 100 million years old. Before then, the universe went through a time called the “dark ages,” when it was almost pitch-black. Space was filled with a featureless gruel, five parts dark matter and one part hydrogen and helium, that thinned out as the universe expanded. Matter was slightly uneven in density, and gravity acted to amplify these density variations: denser regions expanded more slowly than less dense ones did. By 100 million years the densest regions did not merely expand more slowly but actually started to collapse. Such regions contained about one million solar masses of material each. They were the first gravitationally bound objects in the cosmos.

Dark matter accounted for the bulk of their mass but was, as its name suggests, unable to emit or absorb light. So it remained in an extended cloud. Hydrogen and helium gas, on the other hand, emitted light, lost energy and became concentrated in the center of the cloud. Eventually it collapsed all the way down to stars. These first stars were much more massive than today’s—hundreds of solar masses. They lived very short lives before exploding and leaving behind the first heavy elements. Over the next billion years or so the force of gravity assembled these million-solar-mass clouds into the first galaxies.

Radiation emitted from primordial hydrogen clouds, which were greatly redshifted by the expansion, should be detectable by giant arrays of radio antennas with a total collecting area of up to one square kilometer. When built, these arrays will watch as the first generation of stars and galaxies ionize the hydrogen and bring the dark ages to an end.



HUBBLE ULTRA DEEP FIELD, the most sensitive optical image of the cosmos ever made, reveals more than 1,000 galaxies in their early stages of formation.

FAINT GLOW OF A HOT BEGINNING

BEYOND THE DARK AGES is the glow of the hot big bang at a redshift of 1,100. This radiation has been redshifted from visible light (a red-orange glow) beyond even the infrared to microwaves. What we see from that time is a wall of microwave radiation filling the sky—the cosmic microwave background radiation

(CMB), discovered in 1964 by Arno Penzias and Robert Wilson. It provides a glimpse of the universe at the tender age of 380,000 years, the period when atoms formed. Before then, the universe was a nearly uniform soup of atomic nuclei, electrons and photons. As it cooled to a temperature of about 3,000 kelvins, the nuclei and electrons came together to form atoms. Photons ceased to scatter off electrons and streamed across space unhindered, revealing the universe at a simpler time before the existence of stars and galaxies.

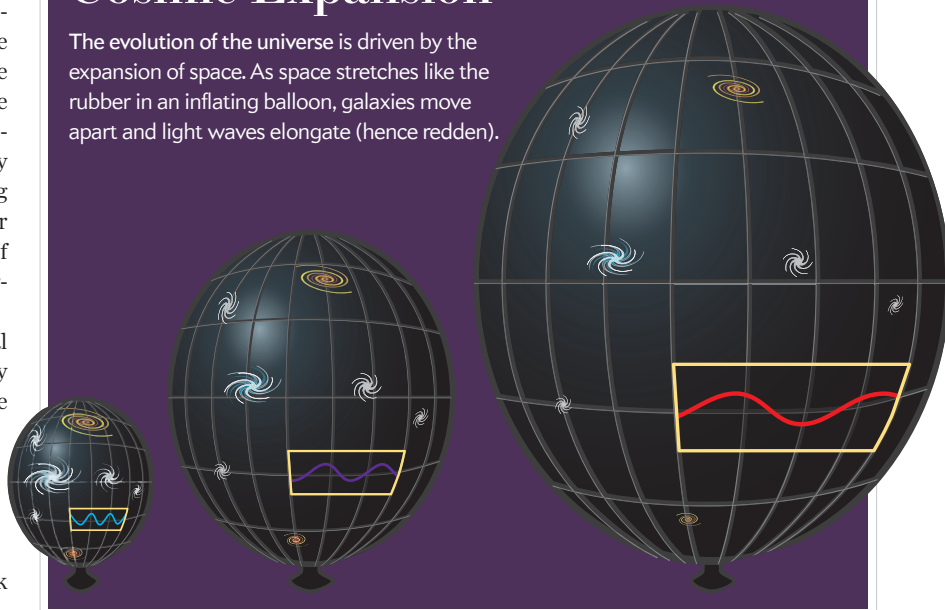
In 1992 NASA’s Cosmic Background Explorer satellite discovered that the intensity of the CMB has slight variations—about 0.001 percent—reflecting a slight lumpiness in the distribution of matter. The degree of primordial lumpiness was enough to act as seeds for the galaxies and larger structures that would later emerge from the action of gravity. The pattern of these variations in the CMB across the sky also encodes basic properties of the universe, such as its overall density and composition, as well as hints about its earliest moments; the careful study of these variations has revealed much about the universe [see *illustration on page 41*].

As we roll a movie of the universe’s evolution back from that point, we see

PRIMER

Cosmic Expansion

The evolution of the universe is driven by the expansion of space. As space stretches like the rubber in an inflating balloon, galaxies move apart and light waves elongate (hence redden).



Earliest Moments of the Big Bang → Formation of Atoms →

The cosmic timeline continues with fairly well-established events leading to the present day.

10^{-35} second

Cosmic inflation creates a large, smooth patch of space filled with lumpy quark soup

10^{-30} s

One potential type of dark matter (axions) is synthesized

10^{-11} s

Matter gains the upper hand over antimatter

10^{-10} s

A second potential type of dark matter (neutralinos) is synthesized

10^{-5} s

Protons and neutrons form from quarks

0.01–300 s

Helium, lithium and heavy hydrogen nuclei form from protons and neutrons

380,000 years

Atoms form from nuclei and electrons, releasing the cosmic microwave background radiation

the primordial plasma becoming ever hotter and denser. Prior to about 100,000 years, the energy density of radiation exceeded that of matter, which kept matter from clumping. Therefore, this time marks the beginning of gravitational assembly of all the structure seen in the universe today. Still further back, when the universe was less than a second old, atomic nuclei had yet to form; only their constituent particles—namely, protons and neutrons—existed. Nuclei emerged when the universe was seconds old and the temperatures and densities were just right for nuclear reactions. This process of big bang nucleosynthesis produced only the lightest elements in the periodic table: a lot of helium (about 25 percent of the atoms in the universe by mass) and smaller amounts of lithium and the isotopes deuterium and helium 3. The rest of the plasma (about 75 percent) stayed in the form of protons that would eventually become hydrogen atoms. All the rest of the elements in the periodic table formed billions of years later in stars and stellar explosions.

Nucleosynthesis theory accurately predicts the abundances of elements and isotopes measured in the most primeval samples of the universe—namely, the oldest stars and high-redshift gas clouds. The abundance of deuterium, which is very sensitive to the density of atoms in the universe, plays a special role: its measured value implies that ordinary matter

amounts to 4.5 ± 0.1 percent of the total energy density. (The remainder is dark matter and dark energy.) This estimate agrees precisely with the composition that has been gleaned from the analysis of the CMB. This correspondence is a great triumph. That these two very different measures, one based on nuclear physics when the universe was a second old and the other based on atomic physics when the universe was 380,000 years old, agree is a strong check not just on our model of how the cosmos evolved but on all of modern physics.

ANSWERS IN THE QUARK SOUP

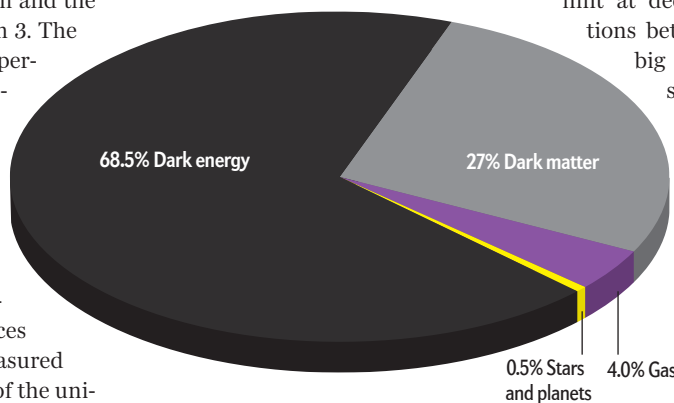
EARLIER THAN A MICROSECOND, even protons and neutrons could not exist and the universe was a soup of nature's basic

building blocks: quarks, leptons, and the force carriers (photons, the W and Z bosons, and gluons). We can be confident that the quark soup existed because experiments at particle accelerators have re-created similar conditions here on Earth today.

To explore this epoch, cosmologists rely not on bigger and better telescopes but also on powerful ideas from particle physics. The development of the Standard Model of particle physics 30 years ago has led to bold speculations, including string theory, about how the seemingly disparate fundamental particles and forces are unified. As it turns out, these new ideas have implications for cosmology that are as important as the original idea of the hot big bang. They hint at deep and unexpected connections between the world of the very

big and of the very small. Answers to three key questions—the nature of dark matter, the asymmetry between matter and antimatter, and the origin of the lumpy quark soup itself—have been starting to emerge.

It now appears that the early quark soup phase was the birthplace of dark matter. The identity of dark matter remains unclear, but its existence is very well established. Our galaxy and every other galaxy, as well as clusters of galaxies, are held together by the gravity of unseen dark matter. Whatever the



BULK OF UNIVERSE consists of dark energy and dark matter, neither of which has been identified. Ordinary matter of the kind that makes up stars, planets and interstellar gas accounts for only a small fraction.

Dark Ages → Modern Era

**380,000–
300 million yr**
Gravity continues to amplify
density differences in the gas
that fills space

300 million yr
First stars and
galaxies form

1 billion yr
Limit of current
observations
(highest-redshift
objects)

3 billion yr
Clusters of galaxies
form; star formation
peaks

9 billion yr
Solar system forms

10 billion yr
Dark energy
takes hold and
expansion begins
to accelerate

13.7 billion yr
Today

dark matter is, it must interact weakly with ordinary matter; otherwise it would have shown itself in other ways. Attempts to find a unifying framework for the forces and particles of nature have led to the prediction of stable or long-lived particles that might constitute dark matter. Some of these hypothetical particles would be present today as remnants of the quark soup phase in the correct numbers to be the dark matter and could even be detected.

One candidate is the called the neutralino, the lightest of a putative new class of particles that are heavier counterparts of the known particles. The neutralino is thought to have a mass between 100 and 1,000 times that of the proton, just within the reach of experiments now under way at the Large Hadron Collider at CERN near Geneva. Physicists have also built ultrasensitive underground detectors, as well as satellite and balloon-borne varieties, to look for this particle or the by-products of its interactions.

A second candidate is the axion, a superlightweight particle about one-trillionth the mass of the electron. Its existence is hinted at by subtleties that the Standard Model predicts in the behavior of quarks. Efforts to detect it exploit the fact that in a very strong magnetic field, an axion can transform into a photon. Both neutralinos and axions

have the important property that they are, in a specific technical sense, “cold.” Although they formed under broiling hot conditions, they were slow-moving and thus easily clumped into galaxies.

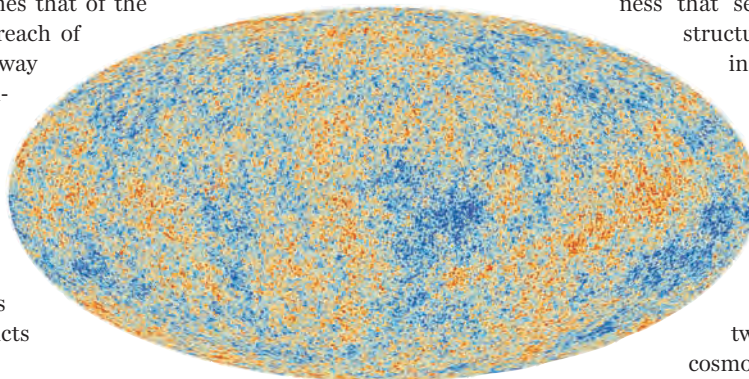
The early quark soup phase probably also holds the secret to why the universe today contains mostly matter rather than both matter and antimatter. Physicists think the universe originally had equal amounts of each, but at some point it developed a slight excess of matter—about one extra quark for every billion antiquarks. This imbalance ensured that enough quarks would survive annihilation with antiquarks as the universe ex-

panded and cooled. More than 40 years ago accelerator experiments revealed that the laws of physics are ever so slightly biased in favor of matter, and in a still to be understood series of particle interactions very early on, this slight bias led to the creation of the quark excess.

The quark soup itself is thought to have arisen at an extremely early time—perhaps 10^{-34} second after the big bang in a burst of cosmic expansion known as inflation. This burst, driven by the energy of a new field (thought to be distantly related to the recently discovered Higgs field) called the inflaton, would explain such basic properties of the cosmos as its general uniformity and the lumpiness that seeded galaxies and other structures in the universe. As the inflaton field decayed away,

it released its remaining energy into quarks and other particles, thereby creating the heat of the big bang and the quark soup itself.

Inflation leads to a profound connection between the quarks and the cosmos: quantum fluctuations in the inflaton field on the subatomic scale get blown up to astrophysical size by the rapid expansion and become the seeds for all the structure we see today. In other words, the pattern seen on the CMB sky is a giant image of the subatomic world. Observations of the CMB agree with this prediction, providing the strongest evidence that inflation or



COSMIC MICROWAVE background radiation is a snapshot of the universe at the tender age of 380,000 years. Tiny variations in the intensity of the radiation (*color-coded here*) are a cosmic Rosetta Stone that reveals key features of the universe, including its age, density, geometry and overall composition.



something like it occurred very early in the history of the universe.

BIRTH OF THE UNIVERSE

AS COSMOLOGISTS TRY to go even further to understand the beginning of the universe itself, our ideas become less firm. Einstein's general theory of relativity has provided the theoretical foundation for a century of progress in our understanding of the evolution of the universe. Because the general theory of relativity does not incorporate quantum theory, the other pillar of contemporary physics, it cannot be relied upon to address the very earliest moments of creation when quantum gravity effects should have been important. The discipline's greatest challenge is to develop a

quantum theory of gravity, with which we will be able to address the so-called Planck era prior to about 10^{-43} second, when spacetime itself was taking shape.

Tentative attempts at a unified theory have led to some remarkable speculations about our very beginnings. String theory, for example, predicts the existence of additional dimensions of space and possibly other universes floating in that larger space. What we call the big bang may have been the collision of our universe with another. The marriage of string theory with the concept of inflation has led to perhaps the boldest idea yet, that of a multiverse—namely, that the universe comprises an infinite number of disconnected pieces, each with its own local laws of physics.

The multiverse concept, which is still in its infancy, turns on two key theoretical findings. First, the equations describing inflation strongly suggest that if inflation happened once, it should happen again and again, with an infinite number of inflationary regions created over time. Nothing can travel between these regions, so they have no effect on one another. Second, string theory suggests that these regions have different physical parameters, such as the number of spatial dimensions and the kinds of stable particles.

The idea of the multiverse provides novel answers to two of the biggest questions in all of science: what happened before the big bang and why the laws of physics are as they are (Albert Einstein's

COSMIC TIMELINE

The Future

Predictable events such as galactic collisions dominate the near future. But the ultimate destiny of our universe hinges on whether dark energy will continue to cause cosmic expansion to accelerate. Broadly, four fates are possible.

a Acceleration ends, and universe expands eternally

100 trillion years
Last stars burn out

b Acceleration continues

30 billion years
Cosmic redout: cosmic acceleration pulls all other galaxies out of our view; all evidence of the big bang is lost

c Acceleration intensifies

50 billion years
Big rip: dark energy tears apart all structures, from superclusters to atoms

20 billion years
Milky Way collides with Andromeda galaxy

d Acceleration changes to rapid deceleration and collapse

30 billion years
Big crunch, perhaps followed by a new big bang in an eternal cycle



(To next cycle)



COSMIC ACCELERATION AND DARK ENERGY

In the Dark

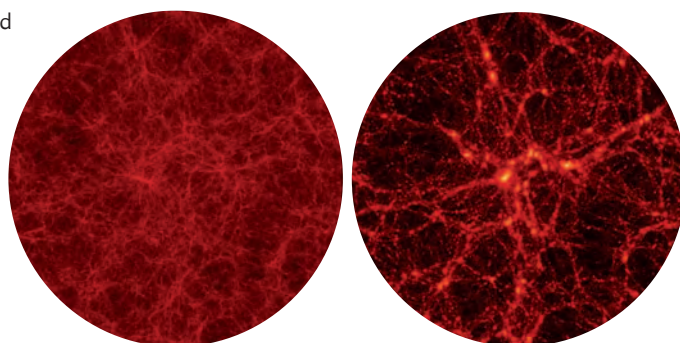
A central feature of our current view of the universe, as well as its biggest mystery, is dark energy, the recently discovered and very weird form of energy that is causing cosmic expansion to speed up. Dark energy took control from matter a few billion years ago. Prior to that, the expansion had been slowing because of the gravitational attraction exerted by matter, and gravity was able to forge structures from galaxies to superclusters. Now, because of the influence of dark energy, structures larger than superclusters cannot form. In fact, had dark energy taken over earlier than it did—say, when the universe was only 100 million years old—structure formation would have ceased before even galaxies could have developed, and we would not be here.

Cosmologists have only rudimentary clues as to what dark energy might be. To speed up expansion of the universe requires a repulsive force, and Einstein's general theory of relativity predicts that the gravity of an extremely elastic form of energy can be actually be repulsive. The quantum energy that fills empty space acts in this way. The trouble is that theoretical estimates of the amount of quantum vacuum energy do not match the amount required by observations of the universe's expansion; in fact, they exceed it by many orders of magnitude. Alternatively, cosmic acceleration might be driven not by a new type of energy but by a process that mimics the repulsive force of such energy, perhaps

the breakdown of general relativity or the influence of unseen spatial dimensions.

If the universe continues to accelerate at the current rate, then in 30 billion years all traces of the big bang will disappear. The light from all but a handful of nearby galaxies will be too redshifted to detect; the temperature of the cosmic background radiation will be too low to measure; and the universe will appear similar to one that astronomers knew 100 years ago before their instruments were powerful enough to reveal the universe we know today.

—M.S.T.



IF THE UNIVERSE had even more dark energy than it does, it would have remained almost formless (*left*), without the large structures that we see (*right*).

famous musing about “whether God had any choice” about the laws). The multiverse makes moot the question of what happened before the big bang because there were an infinite number of big bang beginnings, each triggered by its own burst of inflation. Likewise, Einstein's question is pushed aside: within the infinity of universes, all possibilities for the laws of physics have been tried, so there is no particular reason for the laws that govern our universe.

Cosmologists have mixed feelings about the multiverse. If the disconnected subuniverses are truly incommunicado, we cannot hope to test their existence; they seem to lie beyond the realm of science. Part of me wants to scream, One universe at a time, please! On the other hand, the multiverse solves various conceptual problems. If correct, it will make Hubble's enlargement of the universe by a mere factor of 100 billion and Copernicus's banishment of Earth from the center of the universe in the 16th century seem like small advances in the under-

standing of our place in the cosmos.

Modern cosmology has humbled us. We are made of protons, neutrons and electrons, which together account for only 4.5 percent of the universe, and we exist only because of subtle connections between the very small and the very large. Events guided by the microscopic laws of physics allowed matter to dominate over antimatter, generated the lumpiness that seeded galaxies, filled space with dark matter particles that provide the gravitational infrastructure, and ensured that dark matter could build galaxies before dark energy became significant and the expansion began to accelerate [see box above]. At the same time, cosmology by its very nature is arrogant. The idea that we can understand something as vast in both space and time as our universe is, on the face of it, preposterous. This strange mix of humility and arrogance has gotten us pretty far in the past century in advancing our understanding of the present universe and its origin. I am bullish on

further progress in the coming years, and I firmly believe we are living in a golden age of cosmology. ■

Michael S. Turner pioneered the interdisciplinary union of particle physics, astrophysics and cosmology with his seminal contributions to inflationary theory and cold dark matter theory and his prediction of dark energy (which he also named). He is director of the Kavli Institute for Cosmological Physics and Bruce V. and Diana M. Rauner Distinguished Service Professor at the University of Chicago. He is also president of the American Physical Society, a member of the National Academy of Sciences, and the recipient of the Helen B. Warner Prize for Astronomy and Dannie Heineman Prize for Astrophysics of the American Astronomical Society and the Julius Edgar Lilienfeld Prize of the American Physical Society.

MORE TO EXPLORE

The Early Universe. Edward W. Kolb and Michael S. Turner. Westview Press, 1994.

The Inflationary Universe. Alan Guth. Basic, 1998.

Quarks and the Cosmos. Michael S. Turner in *Science*, Vol. 315, pages 59–61; January 5, 2007.

Dark Energy and the Accelerating Universe. Joshua Frieman, Michael S. Turner and Dragan Huterer in *Annual Reviews of Astronomy and Astrophysics*, Vol. 46, pages 385–432; 2008. <http://arxiv.org/abs/0803.0982>





Explore This Section



Cosmic History

CONTENTS

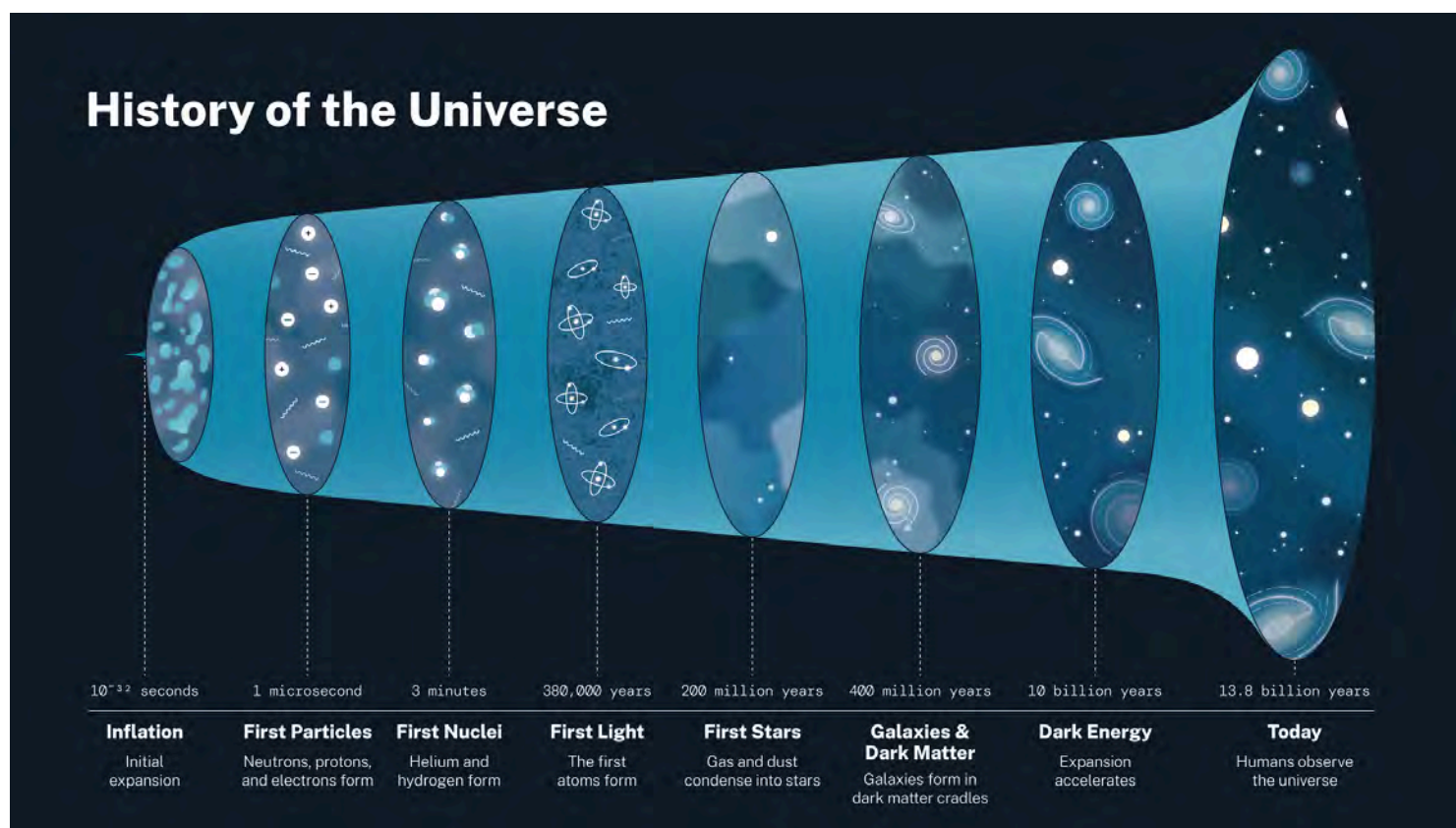
- The Universe's History
- Cosmic Inflation
- Big Bang and Nucleosynthesis
- Recombination
- Dark Ages
- First Stars
- Reionization
- The Future

The Universe's History

The origin, evolution, and nature of the universe have fascinated and confounded humankind for centuries. New ideas and major discoveries made during the 20th century transformed cosmology – the term for the way we conceptualize and study the universe – although much remains unknown. Here is the history of the universe according to cosmologists' current theories.

Cosmic Inflation

Around 13.8 billion years ago, the universe expanded faster than the speed of light for a fraction of a second, a period called cosmic inflation. Scientists aren't sure what came before inflation or what powered it. It's possible that energy during this period was just part of the fabric of space-time. Cosmologists think inflation explains many aspects of the universe we observe today, like its flatness, or lack of curvature, on the largest scales. Inflation may have also magnified density differences that naturally occur on space's smallest, quantum-level scales, which eventually helped form the universe's large-scale structures.



The history of the universe is outlined in this infographic.
NASA

Big Bang and Nucleosynthesis

When cosmic inflation stopped, the energy driving it transferred to matter and light – the big bang. One second after the big bang, the universe consisted of an extremely hot (18 billion degrees Fahrenheit or 10 billion degrees Celsius) primordial soup of light and particles. In the following minutes, an era called nucleosynthesis, protons and neutrons collided and produced the earliest elements – hydrogen, helium, and traces of lithium and beryllium. After five minutes, most of today’s helium had formed, and the universe had expanded and cooled enough that further element formation stopped. At this point, though, the universe was still too hot for the atomic nuclei of these elements to catch electrons and form complete atoms. The cosmos was opaque because a vast number of electrons created a sort of fog that scattered light.

Big Bang News

ARTICLE

6 MIN READ

How NASA’s Roman Space Telescope Will Illuminate Cosmic Dawn

ARTICLE

11 MIN READ

What is Dark Energy? Inside our accelerating, expanding Universe

Inflation

Initial expansion

First Particles

Neutrons, protons, and electrons form

First Nuclei

Helium and hydrogen form

First Light

The first atoms form

First Stars

Gas and dust condense into stars

Galaxies & Dark Matter

Galaxies form in dark matter cradles

Dark Energy

Expansion accelerates

Today

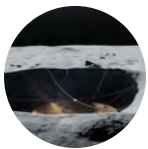
Humans observe the universe



4 MIN READ

NASA's Webb Identifies the Earliest Strands of the Cosmic Web

ARTICLE 1 YEAR AGO



6 MIN READ

Lunar Crater Radio Telescope: Illuminating the Cosmic Dark Ages

ARTICLE 4 YEARS AGO



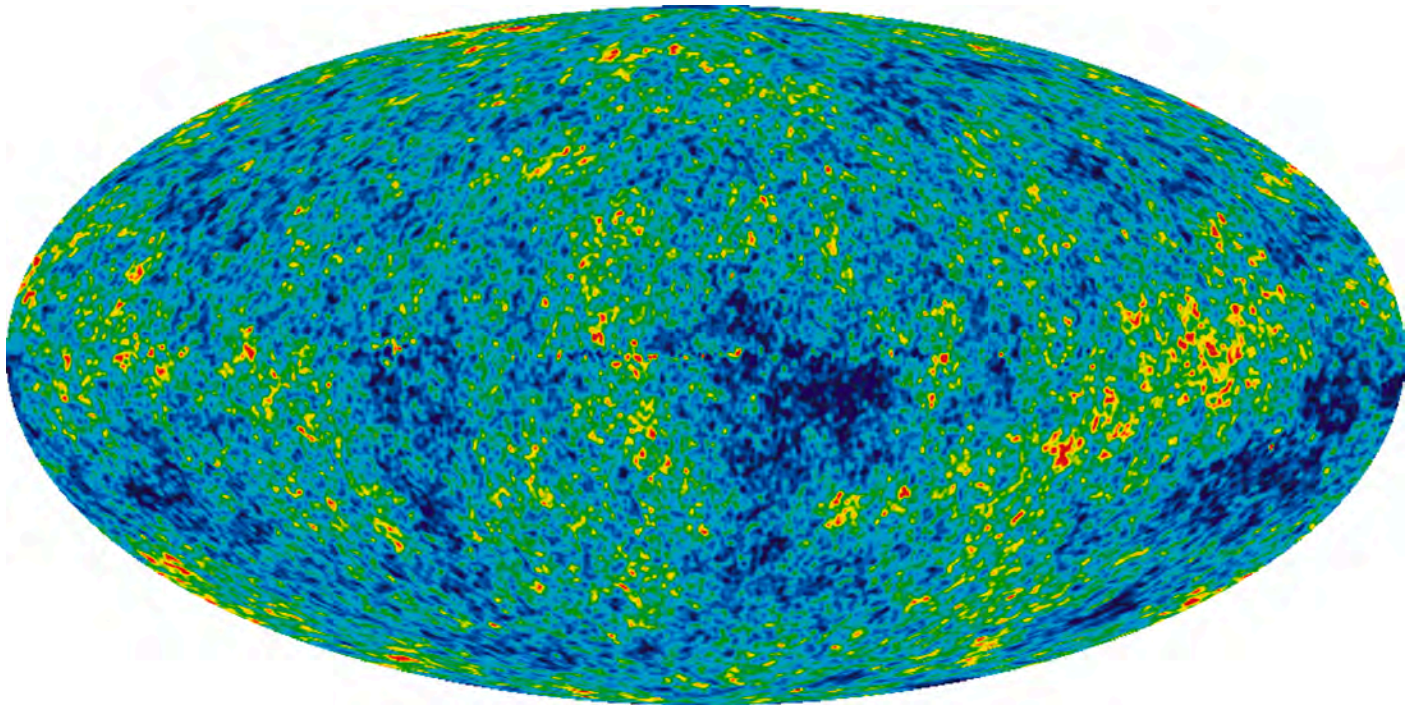
6 MIN READ

Roman Space Telescope Could Image 100 Hubble Ultra Deep Fields at Once

ARTICLE 4 YEARS AGO

Recombination

Around 380,000 years after the big bang, the universe had cooled enough that atomic nuclei could capture electrons, a period astronomers call the epoch of recombination. This had two major effects on the cosmos. First, with most electrons now bound into atoms, there were no longer enough free ones to completely scatter light, and the cosmic fog cleared. The universe became transparent, and for the first time, light could freely travel over great distances. Second, the formation of these first atoms produced its own light. This glow, still detectable today, is called the cosmic microwave background. It is the oldest light we can observe in the universe.



Scientists used nine years of data from NASA's Wilkinson Microwave Anisotropy Probe to create this detailed, all-sky image of the cosmic microwave background. The image reveals 13.8-billion-year-old temperature fluctuations (shown as different colors) – seeds that grew into the galaxies we see today.

NASA/WMAP Science Team

Dark Ages

After the cosmic microwave background, the universe again became opaque at shorter wavelengths due to the absorbing effects of all those hydrogen atoms. For the next 200 million years the universe remained dark. There were no stars to shine. The cosmos at this point consisted of a sea of hydrogen atoms, helium, and trace amounts of heavier elements.

First Stars

Gas was not uniformly distributed throughout the universe. Cooler areas of space were lumpier, with denser clouds of gas. As these clumps grew more massive, their gravity attracted additional matter. As they became denser, and more compact, the centers of these clumps became hotter – hot enough eventually that nuclear fusion occurred in their centers. These were the first stars. They were 30 to 300 times more massive than our Sun and millions of times brighter. Over several hundred million years, the first stars collected into the first galaxies.

Reionization

At first, starlight couldn't travel far because it was scattered by the relatively dense gas surrounding the first stars. Gradually, the ultraviolet light emitted by these stars broke down, or ionized, hydrogen atoms in the gas into their constituent electrons and protons. As this reionization progressed, starlight traveled farther, breaking up more and more hydrogen atoms. By the time the universe was 1 billion years old, stars and galaxies had transformed nearly all this gas, making the universe transparent to light as we see it today.

The Future

For many years, scientists thought the universe's current expansion was slowing down. But in fact, cosmic expansion is speeding up. In 1998, astronomers found that certain supernovae, bright stellar explosions, were fainter than expected. They concluded this could only happen if the supernovae had moved farther away, at a faster rate than predicted.

Scientists suspect a mysterious substance they call dark energy is accelerating expansion. Future research may yield new surprises, but cosmologists suggest it's likely the universe will continue to expand forever.

KEEP EXPLORING

Discover More Topics From NASA

[Galaxies](#) →[Black Holes](#) →[Stars](#) →

National Aeronautics and Space Administration

NASA explores the unknown in air and space, innovates for the benefit of humanity, and inspires the world through discovery.

[About NASA's Mission](#)

[Join Us](#) →

[Home](#)

[News & Events](#)

[Multimedia](#)

[NASA+](#) LIVE

[Missions](#)

[Humans in Space](#)

[Earth & Climate](#)

[The Solar System](#)

[The Universe](#)

[Science](#)

[Aeronautics](#)

[Technology](#)

[Learning Resources](#)

[About NASA](#)

[NASA en Español](#)

Follow NASA

[Sitemap](#) [For Media](#) [Privacy Policy](#) [FOIA](#) [No FEAR Act](#)

[Office of the IG](#) [Budget & Annual Reports](#) [Agency Financial Reports](#)

[More NASA Social Accounts](#)

[NASA Newsletters](#)

[Contact NASA](#)

[Accessibility](#)

Page Last Updated: **9 months ago**

Page Editor: **SMD Content Editors**

Responsible NASA Official for Science: **Dana Bolles**

G1.2 Big Bang Theory

The Big Bang is the leading theory of how the universe came into existence. It derives from a combination of astronomical observations and theoretical models. The concepts behind the Big Bang are not very easy to understand, and in fact are more than a little strange!

In this lesson, you will view a video lecture about the Big Bang, followed by readings that go into greater detail on this theory. Watch the video below. Remember: You can stop and rewind it and replay it as many times as you need to understand the main points. When you are confident that you could explain the main points of the video, continue your study of the Big Bang theory by reading the detailed information in the sources linked below.

When you have completed this lesson, you should be able to summarize the important points of the Big Bang theory in your own words.

View

Big_bang_video



Reading

1. [M. Turner. "Origin of the Universe," *Scientific American* \(2013\).](https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?wrap=1)
 [\(https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?wrap=1\)](https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?wrap=1) 
 [\(https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?download_frd=1\)](https://catcourses.ucmerced.edu/courses/32116/files/7102568/download?download_frd=1)

2. **NASA Astrophysics: The Big Bang**  (<http://science.nasa.gov/astrophysics/focus-areas/what-powered-the-big-bang/>).
-

2

[<-Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-1-origins-of-the-universe)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check)

So what is the big bang theory for the origin of the universe?

The big bang theory postulates that our current universe, and all matter and energy in it, was once contained in a single point in space

and that it expanded very rapidly into existence

at the instant at the Big Bang. Now, this theory is not very easy to understand and it's based on both astronomical observations and mathematical theory.

So, his short clip from National Geographic with couple expert views on the Big Bang.

The Big Bang is an amazingly difficult thing to wrap your brain around, which isn't surprising when you're trying to think about the universe being smaller than the head of a pin.

Actually, smaller than an atom! About as small

as the proton at the center of an atom. Whatever the universe was

at the very beginning some 13.7 billion years ago,

we think it was tiny an infinitesimal nugget of space.

And then, something happened, triggering the most colossal explosion in history.

In a billionth of a billionth

of a billionth of a second, the universe grew from the size of that tiny point to something billions of miles across.

This colossal blast created everything,

all the matter we see in the Universe today

and it did all this

without even making a sound.

When one sees the Big Bang portrayed

in a film there's always a "big bang," a noise. Of course, that's silly because sound doesn't propagate in space; but, what's often overlooked is that

the Big Bang is not just an explosion that happened at some moment in time, at some point in space.

It was not just an explosion of material into

an empty universe that was sitting there waiting to be filled it was the explosion of

space and time into existence to begin with.

And that's what trips most of us up.

Before the Big Bang there was nothing,

literally nothing. And the black void of space is something. So is time.

To ask what's before the Big Bang, what's before there is time,

is really meaningless question, and likewise to ask where the Big Bang occurred is meaningless because there was no space.

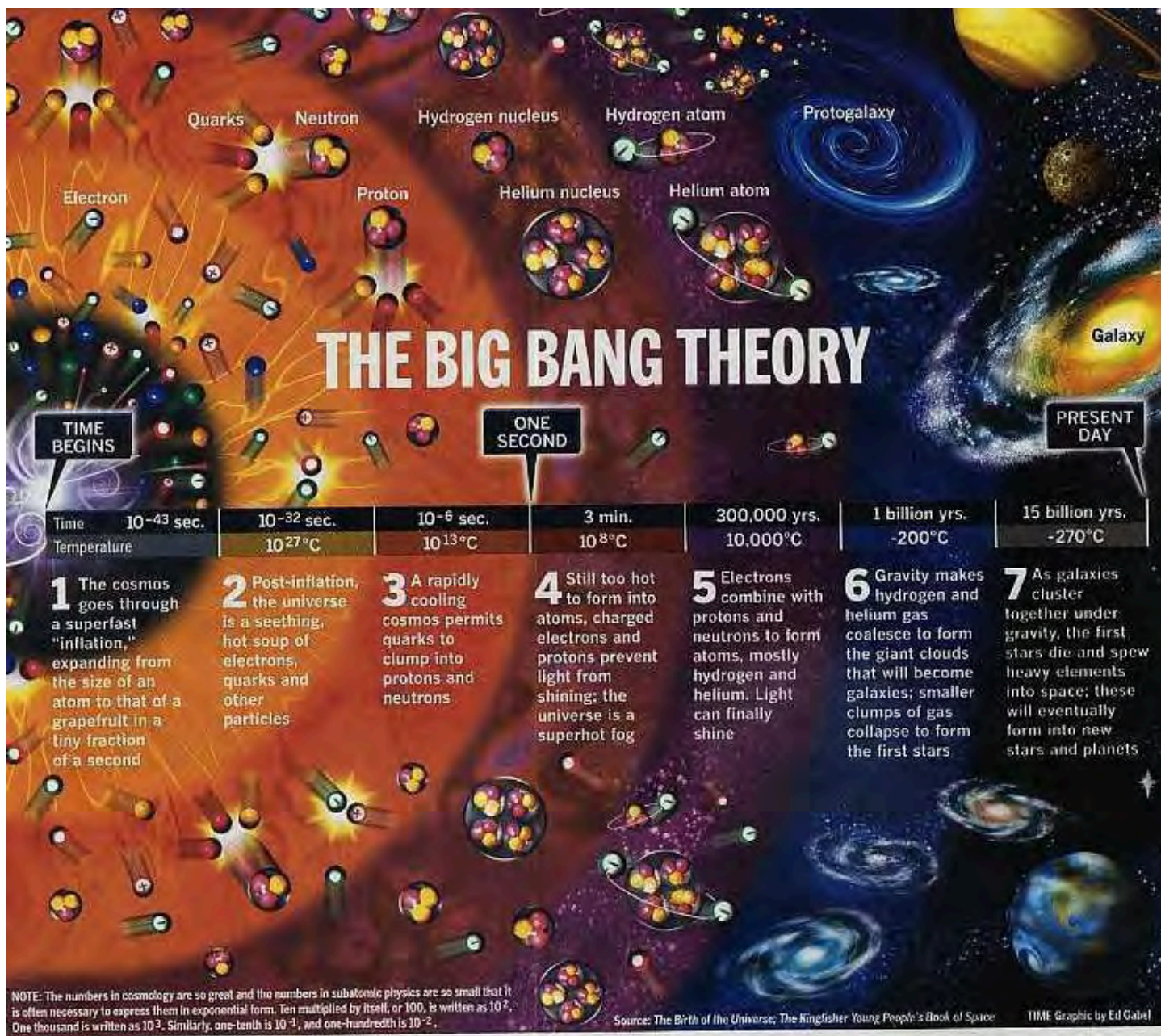
Okay, so there's the expert opinion. It's important to remember that the Big Bang was not an explosion, but rather this instantaneous appearance of

time and space everywhere in the universe. The theory doesn't explain what

might have existed before the Big Bang,
what caused it or what the universe is expanding into.
The age of the universe, based on the Big Bang Theory,
comes from measuring its observable size.
Basically, how far away are the furthest galaxies?
Due to the finite speed of light
and the size of our observable universe, if all matter in energy
condensed back to a single point in time and space,
it's estimated that the Big Bang occurred
about 13.7 billion years ago. Keep in mind that this
estimate might change in the future with new astronomical
observations.

G1.3 Big Bang Theory Self-Check

Review



Key Points: The Big Bang Theory for the Origin of the Universe

1. The Big Bang was not an explosion, but the nearly instantaneous appearance of space and time in the universe;
2. The Big Bang does not explain what existed before, what the universe is expanding into, or what gave rise to the Big Bang;

3. The estimate for the age of the universe is based on the finite speed of light and the size of the universe that we currently observe.
-

3

[<- Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-2-big-bang-theory\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-2-big-bang-theory)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-4-evidence-supporting-the-big-bang-theory\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-4-evidence-supporting-the-big-bang-theory)

G1.4 Evidence Supporting the Big Bang Theory



NASA Images from the Hubble Telescope, named after Edwin Hubble

One of the most important discoveries of 20th century astronomy was made in 1929 by Edwin Hubble (for whom the Hubble space telescope was named). Using astronomical observations, he showed that galaxies generally have Doppler shifts towards the red end of the electromagnetic (EM) spectrum, which indicates that Earth and the source of the light – the galaxies – are moving away from each other.

What is a Doppler shift?

What does Hubble's observation, a Doppler shift towards the red end of the EM spectrum, mean?

In order to explain Hubble's observation, we need to review a little about energy and the electromagnetic spectrum...

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-3-big-bang-theory-self-check)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-5-what-is-energy\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-5-what-is-energy)

G1.5 What is Energy?

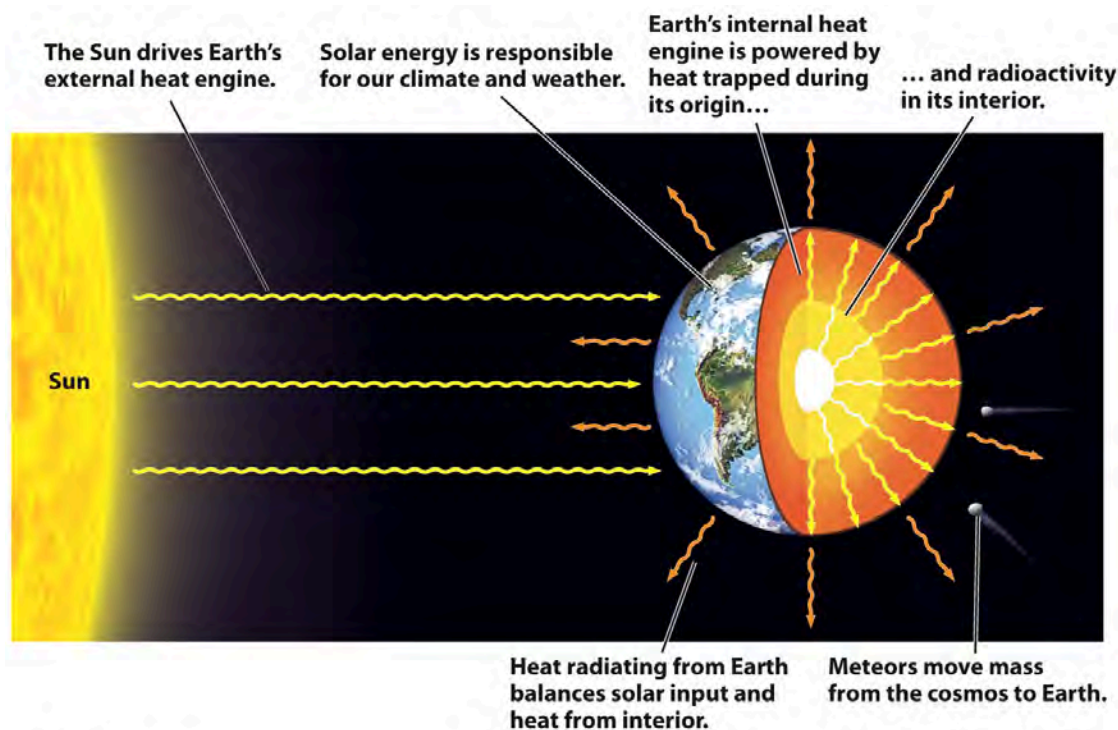


Figure 1.13
Understanding Earth, Sixth Edition
© 2010 W. H. Freeman and Company

Energy exists in different forms and enables different kinds of phenomena to take place. In thermodynamics, energy is defined as "the ability to do work." "Work" includes moving objects, causing substances to react with each other or change form, and transferring heat.

Energy can be described in one of two forms:

1. Potential energy:
Energy that is stored in a system
2. Kinetic energy:
Energy that is expressed as motion

All processes in the Earth's systems are driven by energy in different forms.

5

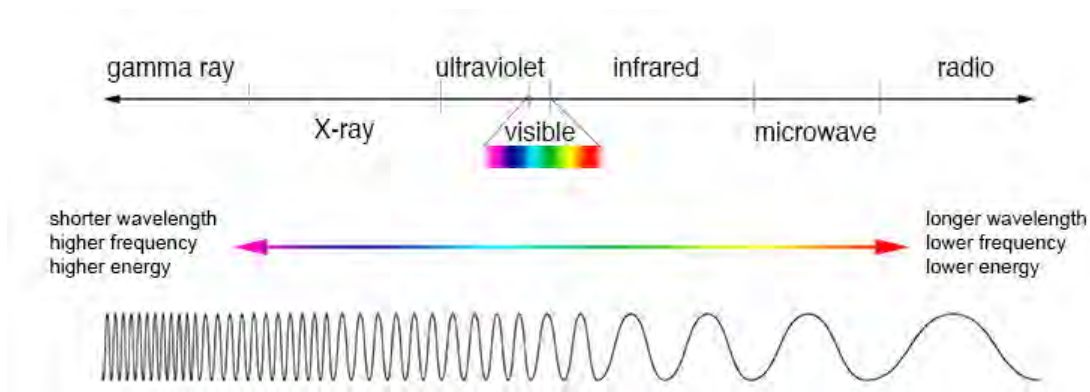
[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-4-evidence-supporting-the-big-bang-theory\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-4-evidence-supporting-the-big-bang-theory)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-6-the-range-of-energy-the-electromagnetic-spectrum\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-6-the-range-of-energy-the-electromagnetic-spectrum)

G1.6 The Range of Energy: The Electromagnetic Spectrum

Schematic diagram, Electromagnetic spectrum (NASA):



The electromagnetic spectrum describes the range of wavelengths or frequencies over which electromagnetic radiation exists. Depending on the form, radiation may be described by different properties and units that are related and interchangeable, such as wavelength, frequency, or energy.

Note that visible light is only a small portion of the entire

electromagnetic
spectrum!

Read on...

6

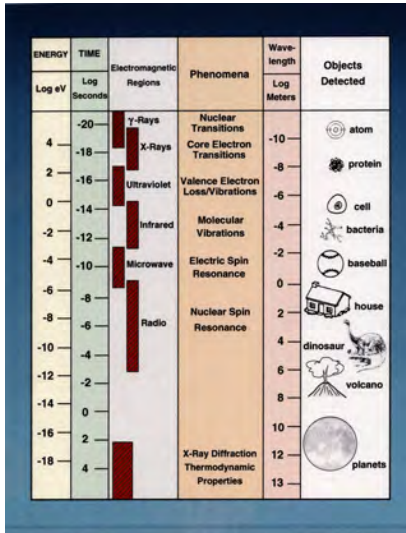
[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-5-what-is-energy\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-5-what-is-energy)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-7-the-electromagnetic-spectrum-continued\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-7-the-electromagnetic-spectrum-continued)

G1.7 The Electromagnetic Spectrum (continued)

The Range of Energy: The Electromagnetic Spectrum (continued)



Energy interconversions are given by simple relationships that allow us to equate energy, distance, and time:

$$E = h\nu$$

or

$$E = \frac{hc}{\lambda}$$

where

E = energy (Joules (J))

h = Planck's constant (J*seconds)

ν = frequency (seconds⁻¹)

λ = wavelength (meters)

c = speed of light

Note the inverse relationship between frequency and wavelength – radiation with a high frequency has a high energy and a short wavelength.

Energy equivalence in space and time allows us to use energy of different wavelengths (or frequencies) to detect objects of different size.

For more information and a chart of unit conversions, visit:

[NASA's website on the Electromagnetic Spectrum](http://imagine.gsfc.nasa.gov/science/toolbox/emspectrum1.html) 
(<http://imagine.gsfc.nasa.gov/science/toolbox/emspectrum1.html>)

7

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-6-the-range-of-energy-the-electromagnetic-spectrum\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-6-the-range-of-energy-the-electromagnetic-spectrum)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-8-doppler-shift-and-hubbles-law\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-8-doppler-shift-and-hubbles-law)



More on Hubble's Law

Looking at the skies from Mt. Palomar in the 1920s, Edwin Hubble found that nearly all far away spiral nebulae (now known as galaxies) in the Universe are moving away from us. This motion, called their recession velocity, is greater the further they are from us. In fact, he found the relationship between a galaxy's velocity away from us (v) and its distance from us (d) approaches a fairly linear one, which is known as Hubble's Law:

$$v = H_0 \times d$$

where H_0 is an observationally determined constant called Hubble's constant. Finding the value of Hubble's constant is still a current topic in astronomy, and it has implications for our understanding of how the Universe evolved since the Big Bang.

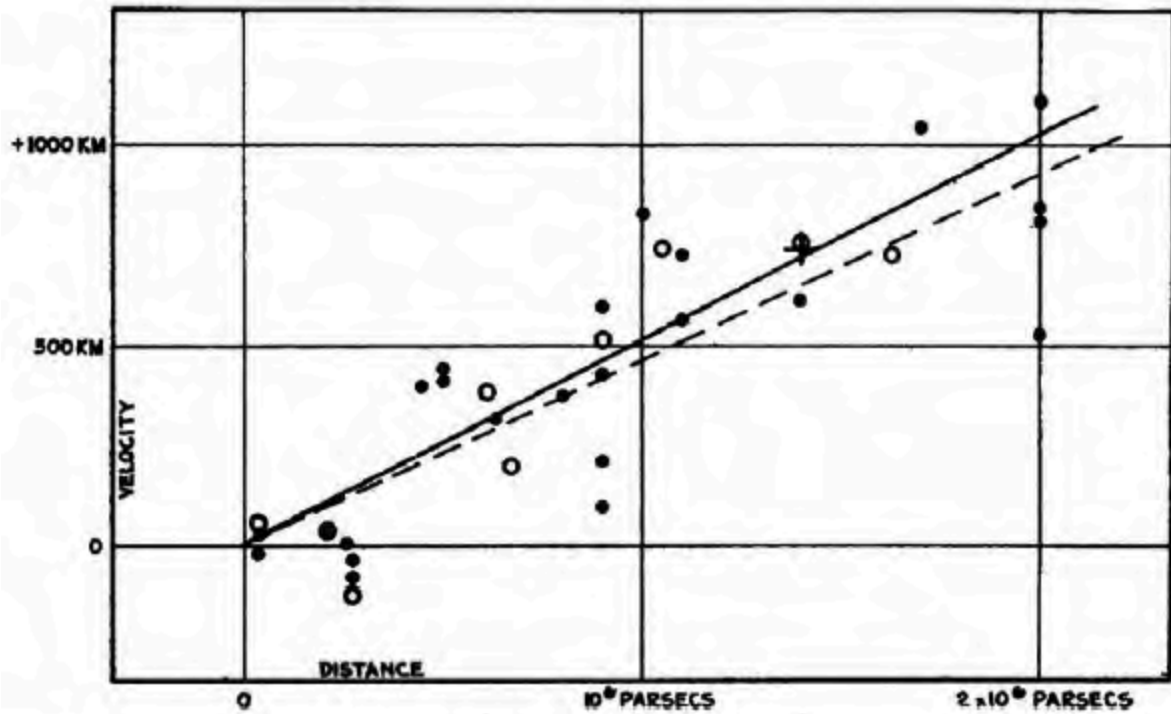
Hubble's finding built on the work of Vesto M. Slipher who, in 1912, discovered that light from nearly all of the "spiral nebulae" he observed, regardless of the direction he looked, appeared to be redshifted. This meant they were moving away.

In 1924 Hubble discovered that the "spiral nebulae" were actually galaxies outside our Milky Way galaxy. Hubble was able, along with his research assistant Milton Humason, to measure distances to a number of the spiral nebulae.

Edwin Hubble sought to find a relationship between their distance from us and their speed. He plotted recessional velocity determined by the doppler shift of stellar spectra as a function distance and established what is now know as Hubble's Law. Hubble's law applies to the objects in the Universe on the largest scales, where the force driving their motions is the expansion of the Universe.



Edwin Hubble
(Image courtesy
AIP Emilio Segre
Visual Archives)



Hubble's graph of redshift versus distance.
(Hubble, Proceedings of the National Academy of Sciences, 1929, 15, 168)



Learn more about what Hubble's Law means



Quiz your knowledge of what Hubble found



Return to solving this using Hubble's Law



Return to the beginning and try another approach

[Home](#)

[Science](#)

[Observatories](#)

[Special Exhibits](#)

[Educators](#)

[Ask An Astrophysicist](#)

Resources

Search

A service of the High Energy Astrophysics Science Archive Research Center (HEASARC), Dr. Andy Ptak (Director), within the Astrophysics Science Division (ASD) at NASA/GSFC

The Imagine Team

Project Leader: Dr. Barbara Mattson

Curator: J.D. Myers

NASA Official: Dr. Andy Ptak

- › [Privacy Policy & Important Notices](#)
- › [Contact NASA](#)
- › [Contact Us](#)
- › [Page Last Updated: Thu, Sep 24, 2020](#)

One of the most important discoveries of 20th century astronomy was made in the 1920s by Edwin Hubble, for whom the Hubble Space Telescope was named. Using astronomical observations, he showed that galaxies generally have Doppler shifts towards the red end of the electromagnetic spectrum.

So, what does that mean?

What is a Doppler shift?

And what does it have to do with the Big Bang theory?

Well, here's a short video from YouTube that gives a really good explanation of Doppler shift and how it relates to Hubble's observations and the Big Bang.

Doppler shift

>> In this podcast, I'm hoping to talk about red shift and the Doppler effect, and how this relates to models of the universe expanding and the theory that the universe began with the Big Bang 14 billion years ago.

But first, the Doppler effect or Doppler shift.

This was first named after the Austrian physicist Christian Doppler, who proposed it in 1842.

And what this states is that there is a change in frequency of a wave for an observer when there is movement relative to the source of the wave.

And this can commonly be seen when a vehicle, producing a siren or horn, approaches, passes, and moves away from an observer.

For example, if you were standing on a street and an ambulance approached and passed you.

First of all, as the ambulance approached you, the sound waves produced in front of the siren will be compressed together as the ambulance moves toward you.

This compression of the sound waves gives a higher frequency or high-pitched sound.

As the ambulance moves past you, you would hear the sound exactly as it's produced.

As the ambulance moves away from you, the sound waves from behind the ambulance are stretched out, giving a low frequency sound or low pitch.

The same thing can be seen with light waves and particularly light waves produced by stars.

For an observer, if a star was moving towards them,

the light waves produced by the star in front of

the star would be compressed together, giving a high frequency light waves.

High frequency visible light is of a blue color, so we call this blueshift.

If a star is moving away from an observer,

the light waves produced behind the star would be stretched out.

And this would give low frequency light waves, and

low frequency visible light is of a red color.

This can be seen with stars when we look out in the universe,

that the stars have a red tinge, or red shift as we call it,

showing that all the stars are moving away from us.

This indicates that the space between stars is getting bigger and therefore the universe is getting bigger.

The further away the stars are, the bigger we see this red shift, so

the universe must be continuing to expand.

Physicists have theorized that the universe is getting bigger now,

and we rewind, we can take the universe back as it shrinks and

shrinks to 14 billion years ago when the universe must have been very, very small.

And by using this,

they've come up with the theory of the Big Bang being the beginning of the universe.

I hope this podcast has helped a bit to explain red shift and

the Doppler effect and how they're used as a piece of

evidence that the Big Bang was the beginning of the universe.

Thank you.

Hubbles law

>> So, Hubble's observation of a red shift in

the light coming from galaxies means that the Earth and

the source of the light, the galaxies, are all moving away from each other.

He also observed that the more distant galaxies were

receding faster than the ones closer to Earth.

These findings are captured in this very simple relationship, which we

now call Hubble's Law, shown here, which states that the receding velocity of

the galaxy is proportional to the distance from the Earth by a constant value,

a value that we've named the Hubble constant.

If all galaxies are moving away from the Earth, that means that

the universe must be expanding in agreement with the Big Bang theory.

Be sure to review the concepts of this lesson and practice solving some

simple problems with Doppler shift, the electromagnetic spectrum and Hubble's Law.

English

AllFrom UC OnlinePresentationsRelatedFor youRecently uploadedWatched

G1.8 Doppler Shift and Hubble's Law



This video [4:20] is about Doppler shift and how Hubble's astronomical calculations support the Big Bang theory.

For more in-depth information, visit [NASA's page on Hubble's Law](http://imagine.gsfc.nasa.gov/features/yba/M31_velocity/hubble_law/more.html)  (http://imagine.gsfc.nasa.gov/features/yba/M31_velocity/hubble_law/more.html)

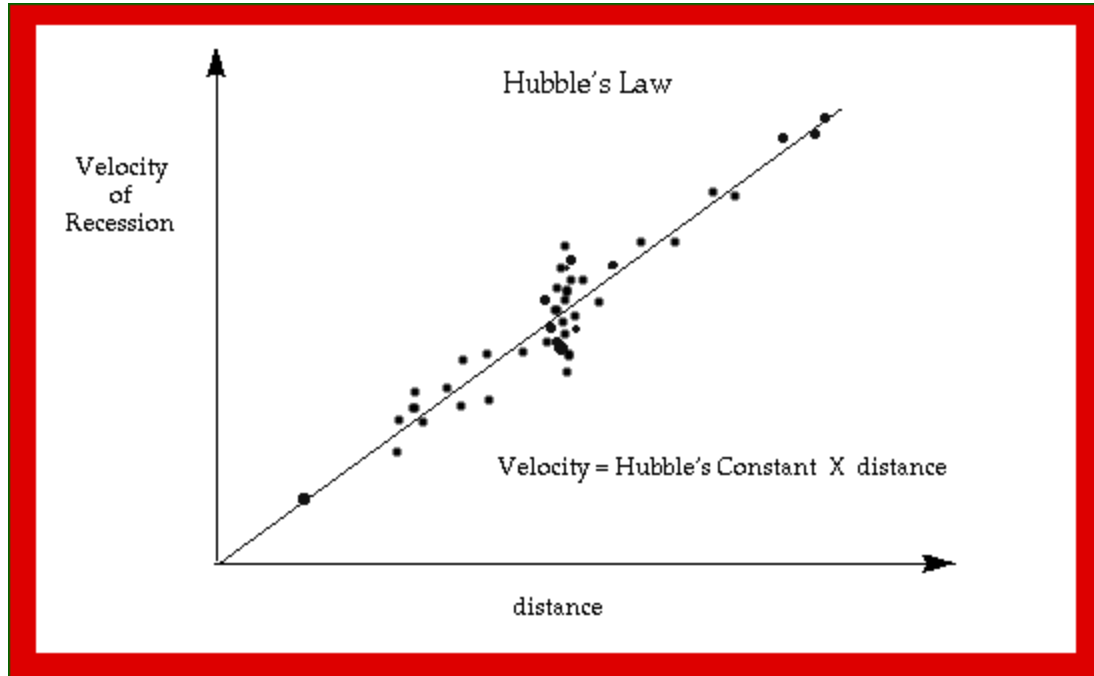
8

[<- Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-7-the-electromagnetic-spectrum-continued\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-7-the-electromagnetic-spectrum-continued)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-9-hubbles-law-and-the-slope-of-a-straight-line\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-9-hubbles-law-and-the-slope-of-a-straight-line)

G1.9 Hubble's Law and the Slope of a Straight Line



Here is an example from the NASA web page illustrating data that supports Hubble's observations.
Note that real data are often messy!

This concludes the first eBook chapter part, *Origins of the Universe*.

[Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-8-doppler-shift-and-hubbles-law\)](https://catcourses.ucmerced.edu/courses/32116/pages/g1-dot-8-doppler-shift-and-hubbles-law)

[Geo1 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-universe)

Geo1: Origins of the Solar System



Part 2: Origins of the solar system & Earth

When you have completed studying Part 1 of this eBook, you should be able to answer these 2 questions:

1. What are the main features of the nebular hypothesis for the origin of the solar system?
2. What was the importance and consequence of the Moon-forming impact during Earth's early history?

Table of Contents

1-2 Origin of the Solar System

(<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-1>) 3-4 Review: Formation of the Solar System (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-3>)

5 Dev. of Our Planetary Neighborhood

(<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-5>) 6 Earth's Early History (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-6>)

7 Review: The Moon-forming Impact (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-7>) (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-8>) 8-9 The Effect of Impacts

(<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-8>)

(<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-11>) 10 Timeline: Early Events in Earth's History (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-10>)

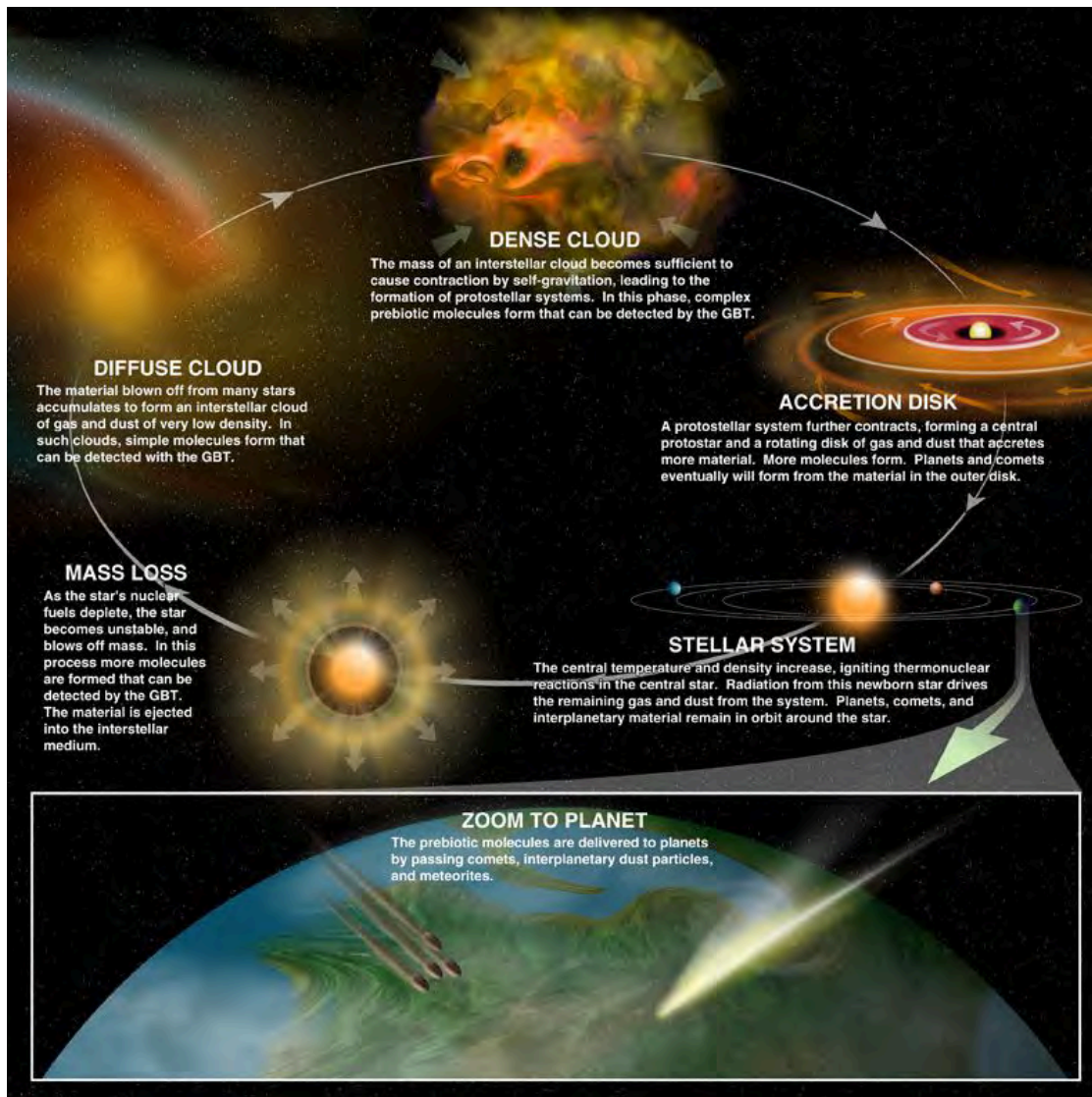
11 Thought Question (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-11>)

12 Summary (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-12>)

Start eBook -> (<https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-1>)

G2.1

Origin of the Solar System: The Nebular Hypothesis



Based on astronomical observations of other stars, our Sun likely formed from a large mass of collapsing gas within an interstellar cloud. The cloud collapsed for millions of years under gravitational forces until it began to rotate.

As gravity pulled the slowly swirling cloud of dust and gas inward, the gas became hotter and denser. Eventually, the temperature and pressure were high

enough that nuclear fusion began and a star was born: the Sun. By the time nuclear fusion began, the cooler outer portions of the solar nebula had become compressed enough to condense solid particles and liquid droplets from the gas. Through accretion, these condensates became the building blocks of the planets, moons, and solid bodies in the solar system.

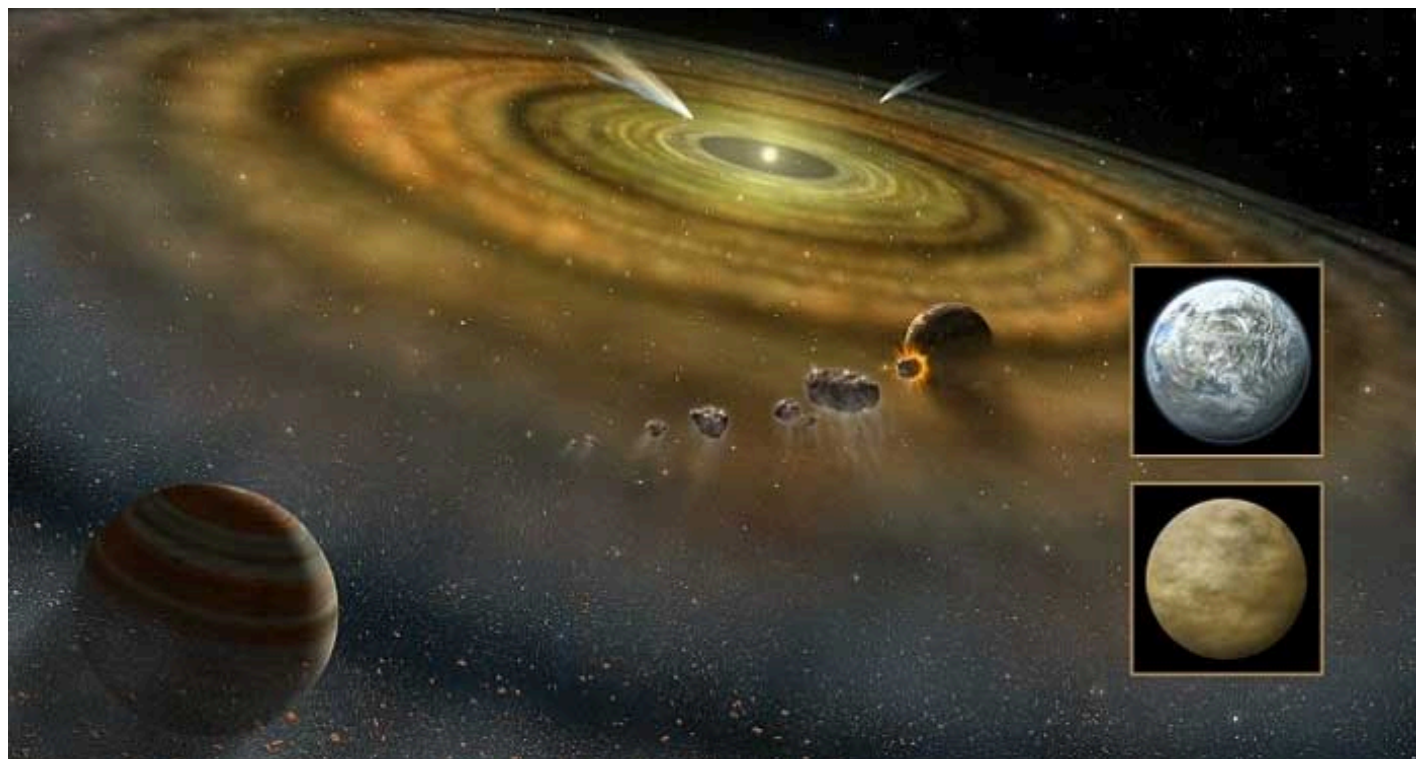
1

[<- Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-0\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-0)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-2\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-2)

Formation of the Solar System: Birth of Worlds



Artist's conception of the dust and gas surrounding a newly formed planetary system. Image Credit: NASA

Like all families, the members of our solar system family share a common origins story. Their story started even before our solar system formed 4.56 billion years ago. Their story started when the story started for every single thing in our [universe](#). Our universe was born from the Big Bang about 13.5 billion years ago. The first stars lived out their lives and eventually exploded, sending "star stuff" out into the cosmos. That original stellar material was recycled as another generation of stars, and many of these, too, exploded at the end of their lives. Our [Sun](#) is thought to be a third-generation star and our entire solar system is made of the recycled star stuff of previous star generations.

Our solar system began forming about 4.6 billion years ago within a concentration of interstellar dust and hydrogen gas called a molecular cloud. The cloud contracted under its own gravity and our proto-Sun formed in the hot dense center. The remainder of the cloud formed a swirling disk called the solar nebula.

Within the solar nebula, scientists believe that dust and ice particles embedded in the gas moved, occasionally colliding and clumping together. Through this process, called "accretion," these microscopic particles formed larger bodies that eventually became planetesimals with sizes up to a few kilometers across. In the inner, hotter part of the solar nebula, planetesimals were composed mostly of silicates and metals. In the outer, cooler portion of the nebula, water ice was the dominant component.



Misconception Alert: Many children and adults have vague ideas of the differences between solar system, galaxy, and universe, and therefore have difficulty distinguishing between the Big Bang from the Universe's

The Sun's light warmed the objects in our solar system, especially those in the inner solar system. There, it was too warm for lightweight volatiles, such as water and ammonia, to condense. In addition, particles from the Sun (the solar wind) pushed volatiles out of the inner solar system. When the volatiles reached the cold temperatures of the outer solar system -- out beyond an invisible boundary called the "frost line" -- they condensed onto the nascent giant planets. Thus, the outer planets had rocks, metals, and volatiles available to accumulate, while the relatively warm, "windy" inner region was stripped of all but the densest materials, like rock and metal. For more information about the compositions of the planets see [A Family Affair](#).

formation, and our own solar system's formation much later.

When the nascent planets grew from a few kilometers to a few hundred kilometers across, they became massive enough that their gravity influenced each other's motions. This increased the frequency of collisions, through which the largest bodies grew most rapidly. During this "childhood" stage of growth, the bodies are referred to as planetesimals. Eventually, regions of the nebula were dominated by large protoplanets. The interiors of these more mature bodies were becoming ordered -- differentiated -- into protoplanets. The process of collision and accretion continued until only four large bodies remained in the inner solar system -- [Mercury](#), [Venus](#), [Earth](#), and [Mars](#), the terrestrial planets. In the cold outer solar nebula, much larger protoplanets formed. The largest ones swept up other protoplanets, planetesimals, and nebular gas, leading to the formation of [Jupiter](#), [Saturn](#), [Uranus](#), and [Neptune](#).

Shortly after Earth formed, the [Moon](#) did. A large object (about half as wide as Earth) collided with our world. The off-center cosmic smash-up increased Earth's spin, and its energy disintegrated the impacting object, melted Earth's outer layers, and flung debris into orbit around Earth. This material formed a ring of gas, dust and molten rock around Earth. In less than a hundred years -- an incredibly short time for the formation of an entire world -- this debris clumped (accreted), growing larger and larger to form our Moon!



Our solar system's childhood was a time of massive violent collisions.

The solar system's birth story is an unfolding tale. The processes of solar nebula collapse and accretion explain why there is so much space in space, where we find the various types of planets and other small bodies, and why the planets all lie in about the same plane and orbit the Sun in the same direction. Despite their diversity, all the planets, dwarf planets, comets, and asteroids in the solar system formed together, along with the Sun, as a system. Scientists still have many questions about our solar system's story. How did the planets form quickly enough to escape the blast of the early Sun's intense solar wind, which would have swept gas and dust out of the growing planets' reach?

Fortunately, vital clues are scattered throughout the solar system -- from the oldest rocks on the Earth, Earth's Moon, Mars and the asteroids to the frozen outer reaches of the Kuiper Belt. NASA robotic missions are examining these distant worlds, making new discoveries that will help to fill in the pages of this story. Ongoing research is examining the early solar nebula's composition and conditions such as radiation, by studying grains from comets, from the Sun, and the composition of Jupiter (for more information about comets, see [Small Bodies / Big Impacts](#)). Comets are made of the ices and dust from the original nebula that formed our solar system, and can tell us more about how our planets formed. The [EPOXI](#) mission, which will fly past Comet Hartley 2 on November 4,

2010, is studying the structure, composition and formation history of cometary nuclei, in order to learn more about the origin of the solar system.

Scientists have examined meteorites to learn more about the primitive material that made up the solar nebula. Recent research on the radioactive isotopes within meteorites suggests a nearby supernova explosion may have influenced the environment and materials in the early solar system, seeding it with materials and perhaps triggering it to collapse into a swirling accretion disk.

Our understanding of the solar system's formation is also being guided by the new worlds discovered around distant stars. The unusual orbits of many of these distant planets have sparked a hot topic of research: that planets' orbits may shift -- migrate -- early after their formation. This "planetary migration" is the best explanation for these newly discovered "hot Jupiters"-- massive gas giants orbiting extremely close to their stars.

Planetary migration is caused by gravitational interactions between the gas in the solar nebula and the young planets, and also by gravitational interactions between the planets and the remaining planetesimals. These interactions transfer angular momentum between the objects, causing the planet too either give up energy, and move closer toward the star, or gain energy, and migrate outward.

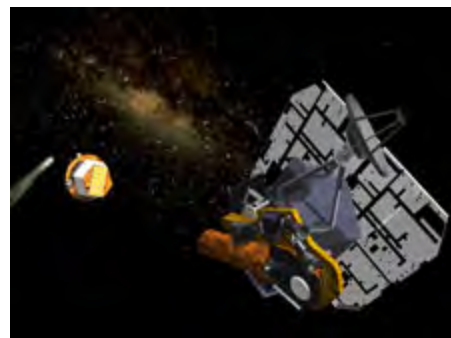
One model for our own solar system suggests that our giant planets' orbits shifted dramatically early in the solar system's history, with Jupiter's orbit migrating slightly inward toward the Sun, and those of Saturn, Neptune, and Uranus expanding farther from the Sun. These dramatic movements gave us the order of the planets and smaller bodies that we are familiar with today, and caused many smaller bodies (such as comets) to scatter out into the Kuiper belt and Oort cloud.

Planetary migration may also account for an intense period of bombardment throughout the inner solar system, around 3.9 billion years ago. Some computer models suggest that interactions of Jupiter and Saturn's orbits destabilized the orbits of asteroids and comets in the outer solar system, causing them to pelt the inner solar system. Scientists are continuing to investigate how the gravitational influences of the giant planets interacted to dramatically reshape our solar system.

Did the planets form in their present locations, or did the giant planets form closer to the Sun and, through complex gravitational interactions, migrate to their orbits of today? More detailed understanding of the dates of impacts and the materials in the early solar system may provide us with final answers.

Further Information:

- [The Outer Planets: How Planets Form](#)
- [PSRD: Triggering the Formation of the Solar System](#)

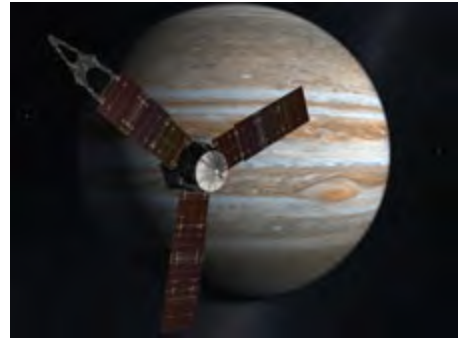


The Deep Impact-EPOXI mission is one of several NASA spacecraft studying comets to find clues about the origin of our solar system.

Featured Missions

[Juno](#), launched in August 2011, will improve our understanding of our solar system's beginnings by revealing the origin and evolution of Jupiter. With its suite of science instruments, Juno will investigate the existence of a solid planetary core, map Jupiter's intense magnetic field, measure the amount of water and ammonia in the deep atmosphere, and observe the planet's auroras. Juno will

let us take a giant step forward in our understanding of how giant planets form and the role these titans played in putting together the rest of the solar system.



Juno is NASA's next mission to Jupiter, our solar system's largest planet.

The most widely held scientific theory of the origin of the Earth and of the entire solar system was first advanced by the German philosopher Immanuel Kant. And later by French mathematician Pierre-Simon Laplace.

They held that the sun and planets condensed from a nebula.

Support for this idea came with development of more powerful telescopes and spectrometers, which can detect the composition of material deep in space.

[NOISE] Nebula are created by the explosion of stars which have exhausted their nuclear fuel.

By studying the compositions of nebula, astronomers have found evidence that stars and planets can form from these interstellar clouds.

99% of nebula material is composed of hydrogen and helium gas.

But about 1% consists of tiny bits of solid matter:

silicon compounds, iron oxides, carbon, and

a host of other materials, all of which are found in the sun and planets.

If the force of gravity between the gas and

bits of cosmic dust is strong enough, this matter is gradually pulled together.

As the nebula contracts,

it starts spinning with heat setting its particles in motion.

The further it condenses, the faster it rotates.

The process is something like what happens when a skater condenses himself by drawing in his arms until his body becomes a blur.

[SOUND] Gradually, the gas and stardust become more and more concentrated at the center of the nebula, heating up to form a star.

The material spinning around a new star flattens into a disk, drawn there by gravity from the equatorial bulge created by the star's rotation.

Over tens of millions of years, dust gathers as clumps in the hot disk.

These collide and weld together under the influence of gravity, forming larger objects called planetesimals, which in turn, come together to form the planets.

The precise details of this accretionary process remain unclear.

Ultimately, the gases themselves partially crystallize and condense to form balls of ice.

>> The cloud of material from which the planets form started out as being mostly gas with a small amount of dust.

That dust was very fine indeed.

But the particles collided with each other and, during those collisions, they stuck with each other.

So that material started to coagulate into still bigger bodies.

So you have to imagine that we went from dust particles all the way up to objects that were kilometers across.

Those objects were colliding with each other, making still bigger bodies.

So that we think of the formation of solid planets like the Earth as being a hierarchical process.

A process that starts off with very small things,

very large in number, towards intermediate size things, smaller in number.

And eventually ending up with a very small number of large objects.
As an intermediate stage in this process,
we could imagine that the region where the Earth formed
contained hundreds of objects that were about the size of the Earth's moon.
Most of those objects then coalesced to make a single planet, the Earth.

English

AllOuter spacePresentationsFor youRecently uploadedWatched

G2.2

Origin of the Solar System: The Nebular Hypothesis

Solar_...



Our solar system formed from giant cloud of mostly hydrogen and helium, and a small percentage of heavier elements. According to the nebular hypothesis (or theory), the planets of the solar system all formed at the same time, from the same material as the Sun, and in a relatively short time, probably less than about 100 million years.

1. View: This PBS Nova video [4:00] explains the Nebular Hypothesis.

2. Read: After you have viewed the video, read:

NASA. **"Formation of the Solar System: Birth of Worlds"** 
(<http://mobile.arc.nasa.gov/public/iexplore/missions/pages/yss/november.html>)

3. Self-Check: When you have completed this page, you should be able to explain the main features of the Nebular Hypothesis for the formation of the solar system.

2

[<- Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-1\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-1)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-3\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-3)

G2.3

Review: Steps in the Formation of the Solar System

Step 1

Figure 9.2 part 1
Understanding Earth, Sixth Edition
© 2010 W. H. Freeman and Company

Step 2

Figure 9.2 part 2
Understanding Earth, Sixth Edition
© 2010 W. H. Freeman and Company

continued on p. 4 ->

3

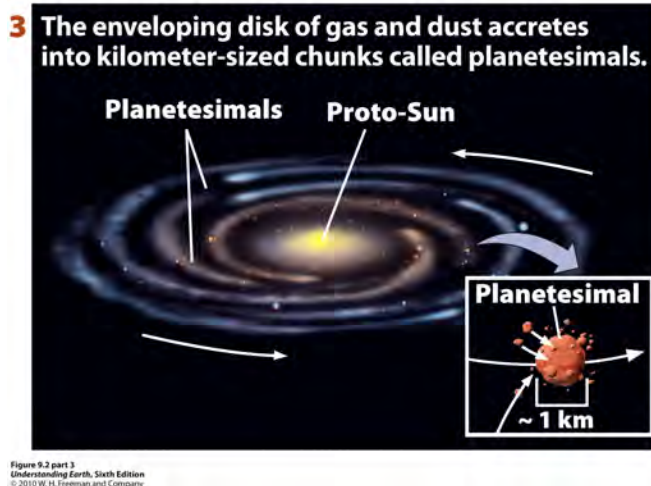
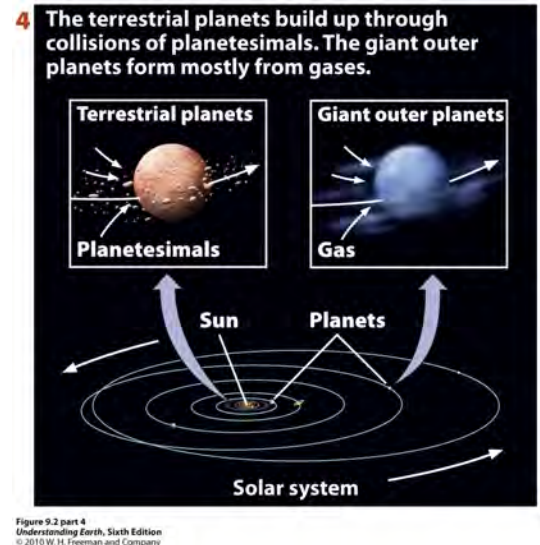
[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-2\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-2)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-4\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-4)

G2.4

Review: Steps in the Formation of the Solar System

Step 3Step 4

4

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-3\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-3)

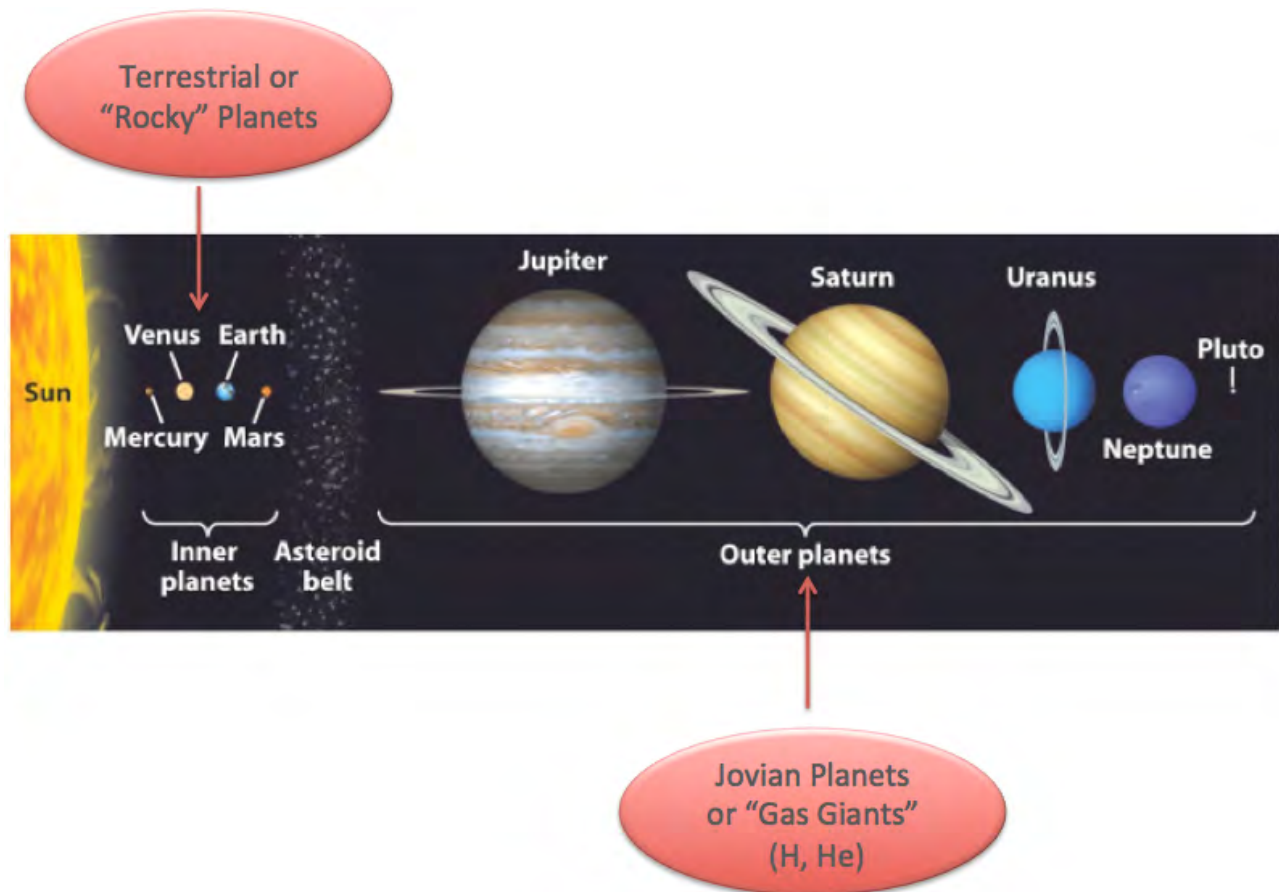
[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-5\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-5)

G2.5

Development of Our Planetary Neighborhood

In the presence of the Sun's gravitational force, the solid bodies in our solar system began to accrete into the planets, moons, and asteroids that we know today. The Sun accounts for 99.85% of the entire mass of the solar system, while the planets make up only about ~0.15% of the mass. Lighter elements, primarily hydrogen (H) and helium (He), condensed to form planets further from the sun, while heavier elements (such as silicon, oxygen, iron, aluminum, and magnesium) were held closer to the Sun and accreted into the inner, "rocky" planets.



But during its early history, something different happened to the Earth that profoundly influenced its evolution...

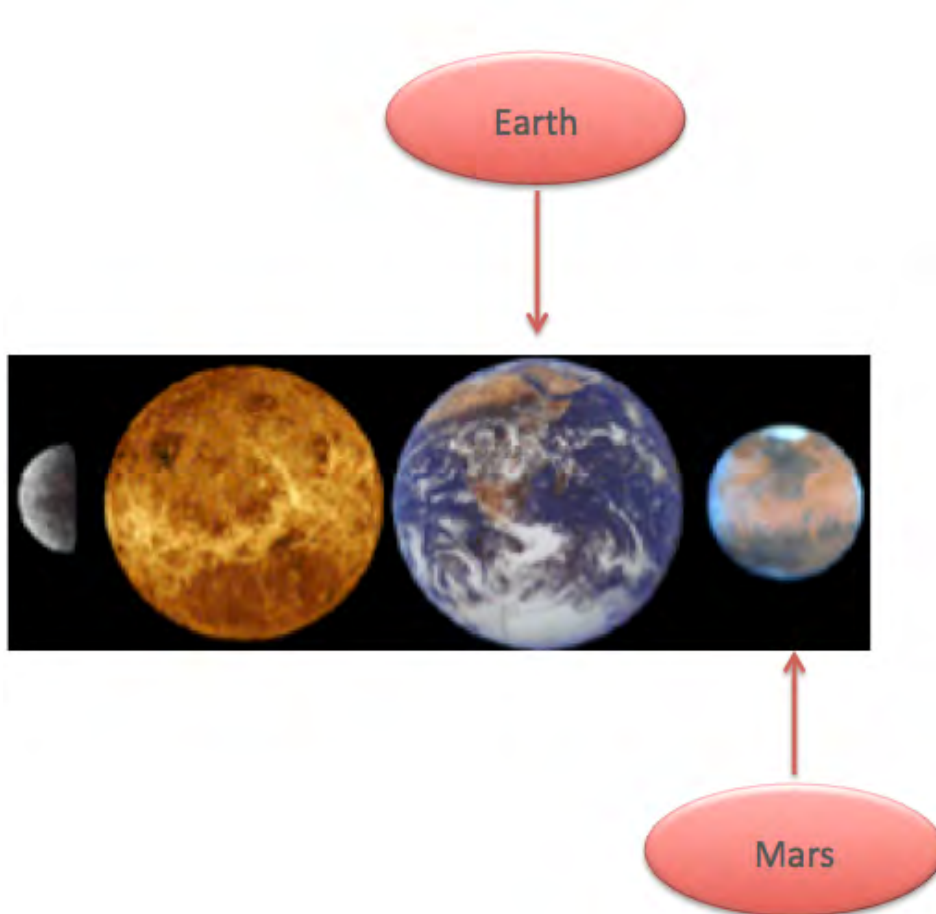
[< - Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-4\).](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-4)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-6\).](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-6)

G2.6

Earth's Early History



Through observations of cratering on other planets, the Earth was subjected to heavy bombardment by planetesimals and larger bodies during its early history. Evidence indicates that the Earth was impacted by a large body about the size of Mars during the later stages of its accretion.

Watch the video below [3:30] to learn more about the Moon-forming impact and the important consequences associated with this event.

Moon-Forming Impact

Collision That Formed The Moon



6

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-5\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-5)

Geo1.2 TOC (<https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system>)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-7\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-7)

the tilt of our planets rotational axis is about 23 degrees currently and it's that tilt that gives us our overall seasonal pattern it turns out that if we didn't have our moon or if we had a moon but it was just smaller than our current moon that interactions with the other planets in the solar system would cause Earth's rotational axis to vary by many tens of degrees on timescales of millions of years this would have profound impact on the overall climate of the earth and while the earth with such a wildly changing tilt might still have been habitable it would certainly have been an entirely different climate on a different earth than we have today and so in this way the formation of our Moon and the formation of our Moon of just the mass of our Moon has had a very important impact on the evolution of the Earth's spin state and with it the evolution of our long-term climate there's a fundamental connection between our studies of the origin of the Moon and the physical samples that were brought back by the Apollo mission the actual rocks from the moon when those rocks were first studied we learned several very significant things one we learned that compositionally those rocks are extraordinarily similar to the upper layers of the earth if you look at an element like oxygen its distribution of different isotopes of oxygen is essentially identical in the moon and the earth whereas it's very different in meteorites from Mars or the asteroid belt another clue another constraint we learned from the Apollo samples is that the moon rocks appeared very dry compared to earth rocks almost as if

they had been heated to a high temperature and that they had lost the elements that tend to vaporize easily upon heating and that's suggested to us that the process that formed the moon must have been a high-energy event so we think the moon formed when the earth was struck by a very large object as the earth was forming so this would have been a separately formed planet that was vying for dominance in the inner solar system along with the earth and other growing planets but it collided with the earth and ultimately was absorbed by the earth and gave birth to our moon so in terms of the impactor which we often call Thea we think that based on its mass that it would have been as big as the planet Mars before it collided with the earth it would have had an iron core and a silicate mantle like the earth and it would have had about 10% of the mass of the earth we think initially the inner solar system had maybe 20 small planets and it was through collisions between these planets that we eventually ended up with our final four terrestrial planets now the last of these big collisions on the earth was the one we think formed the moon the reason that we focus on a mars-sized impactor is that that is the impact or size that if you hit the earth with a mars-sized impactor at an off-center angle at the expected speed you will start the earth spinning with about a five-hour day now that's a very rapid spin rate but it turns out that is what is needed to agree with our current 24-hour day with the moon at its current distance the moon is moving away from the earth so we know when the moon first formed it was much closer to the earth and the earth was spinning much faster

so the idea that the moon formed by this giant collision with the earth was first proposed in the mid 70s and yet we couldn't test the idea initially for smaller scale collisions you can perform experiments in the laboratory but you can't perform an experiment of what happens when two planets collide in any laboratory setting to do this you need to build a computational model to simulate the planets and their response to the collision we can use our computer simulations as an essence a virtual laboratory to test how the outcome depends on things like the size of the impactor the impact speed and angle and use the overall results to assess the likelihood of forming a moon like ours so we're simulating this final large collision with the earth from the perspective of someone on the earth at the time what you would have seen is an enormous object approaching the earth many many many times larger than the full moon in the sky it would have hit the earth at an oblique angle on a timescale of a few hours an enormous shock wave from the impact would have propagated around the entire surface of the earth the vaporized rock from the initial impact point would have been ejected out and it would have flowed around the entire earth [Music] and the earth would find itself enveloped by a thick atmosphere but not like an atmosphere we know today this would have been an atmosphere of vaporized rock with a temperature of more than 5,000 degrees [Music] the momentum force of this collision is enough to start the earth rotating very

rapidly and so only hours after this impact the earth has a rapid rotation rate

now we think on a much longer time scale of about a hundred years that disk will cool the vapor will condense and out of that disk through collisions within the disk the moon would have grown in the community

so although we think we have a good idea about how the moon formed the exact details of this event are still being debated and this is in large part a testament to how those samples that were returned decades ago are still driving active science debate as ongoing analysis provides us with more clues from those samples on how the Earth Moon system formed

G2.7

REVIEW: The Moon-Forming Impact

Based on the nebular hypothesis and radiometric age dating of rocks from the Moon, a current estimate of the timing of the Moon-forming impact is about 4.51 billion years (b.y.) ago, which is about 40-50 million years (m.y.) after the beginning of solar system formation – not very long in geologic time!

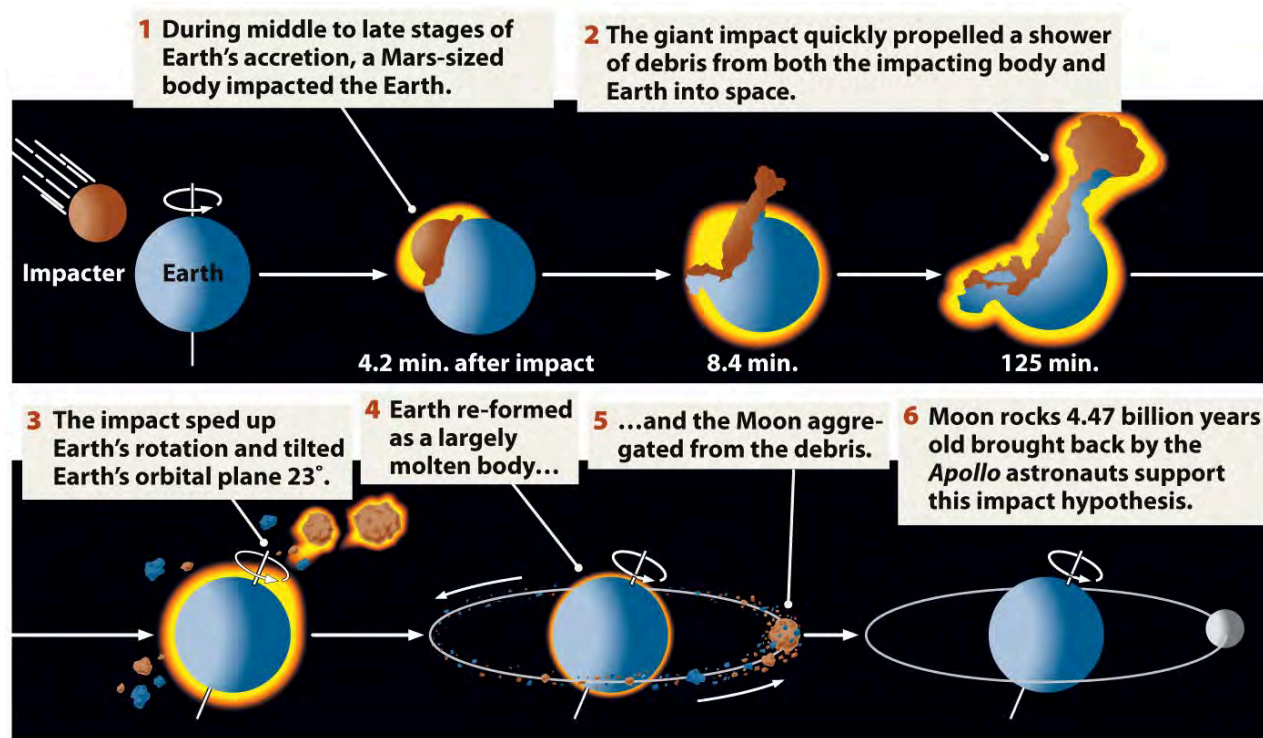


Figure 9.4
Understanding Earth, Sixth Edition
 © 2010 W. H. Freeman and Company

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-6\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-6)

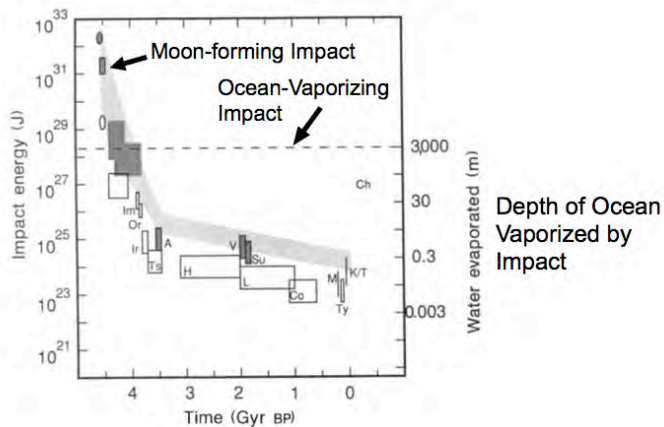
[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-8\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-8)

G2.8

The Effect of Impacts

Impact Energies of the Largest Impacts on the Earth and Moon



Even after the Moon-forming impact and final accretion phase of planets in the solar system, there remained a large amount of residual solid material – meteorites and asteroids – that frequently collided with the planets, the Earth, and the Moon during the early history of the solar system before about 3.9 b.y.

Evidence of these collisions is found in the cratered surface of Mars, our Moon, and other bodies in the solar system. Theoretical models constructed from this cratering history, and sites of known impacts on Earth, are used to estimate the frequency and size of impacts during Earth's geologic history, which is illustrated in the graph.

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-7\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-7)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-9\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-9)

G2.9

The Effect of Impacts

TABLE 9.2 Impacts by Asteroids and Meteorites and Their Effects on Life on Earth

	Example or Size Equivalent	Most Recent	Planetary Effects	Effects on Life
Supercolossal: radius (R) > 2000 km	Moon-forming event	4.51×10^9 years ago	Melts planet	Drives off volatiles; wipes out life on Earth
Colossal: R > 700 km	Pluto	More than 4.3×10^9 years ago	Melts crust	Wipes out life on Earth
Huge: R > 200 km	4 Vesta (large asteroid)	About 4.0×10^9 years ago	Vaporizes oceans	Life may survive below surface
Extra large: R > 70 km	Chiron (largest active comet)	3.8×10^9 years ago	Vaporizes upper 100 m of oceans	May wipe out photosynthesis
Large: R > 30 km	Comet Hale-Bopp	About 2×10^9 years ago	Heats atmosphere and surface to about 1000 K	Burns continents
Medium: R > 10 km	Cretaceous-Tertiary impactor; 433 Eros (largest near-Earth asteroid)	65×10^6 years ago	Causes fires, dust, darkness; chemical changes in ocean and atmosphere; large temperature swings	Cretaceous-Tertiary impact caused extinction of 75 percent of species and all dinosaurs
Small: R > 1 km	About size of near-Earth asteroids	About 300,000 years ago	Causes global dusty atmosphere for months	Interrupts photosynthesis; individuals die but few species extinct; threatens civilization
Very small: R > 100 m	Tunguska event (Siberia)	1908	Knocked over trees of kilometers away; caused minor hemispheric effects; dusty atmosphere	Newspaper headlines; romantic sunsets; increased birth rate

Source: J. D. Lissauer, *Nature* 402 (1998): C11–C14.

Table 9.2
Understanding Earth, Sixth Edition
 © 2010 W. H. Freeman and Company

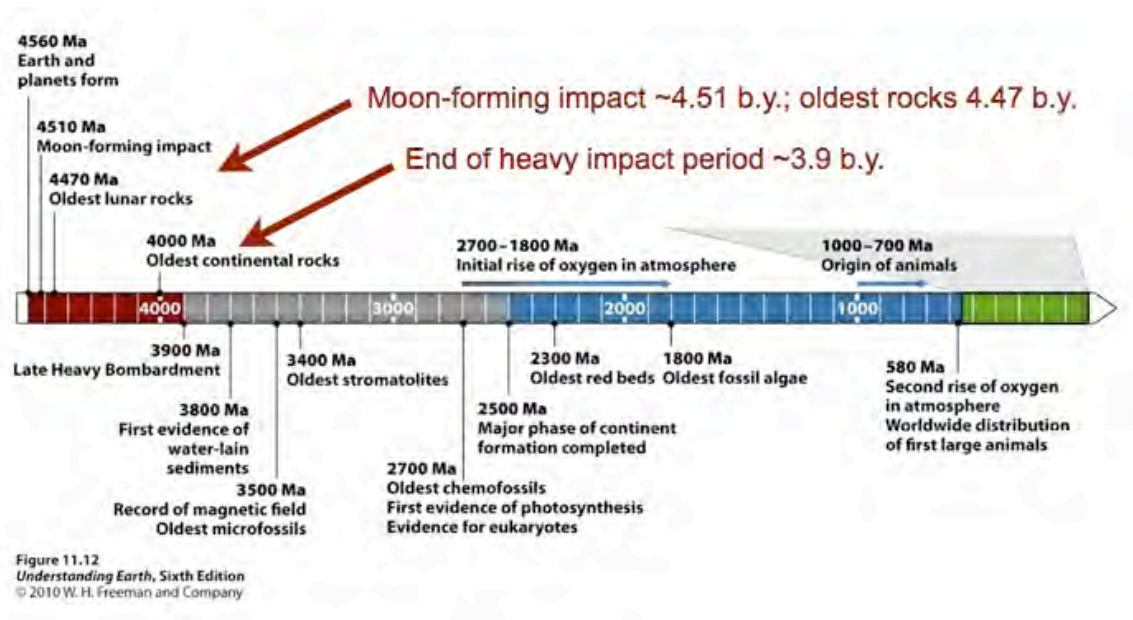
The amount of energy released in the impact of an extra-terrestrial body with Earth depends mostly on the size of the impacting meteorite or asteroid – the larger the body, the more catastrophic the consequences!

This table summarizes predicted effects on Earth for impactors of different radius (R). It is thought that the extinction of the dinosaurs at the end of the Cretaceous period 65 million years ago was caused by a medium-sized impactor, perhaps only about 10 kilometers (km) in diameter.

G2.10

Timeline of Early Events in Earth's History

The Moon-forming impact occurred about 4.51 billion years (b.y.) ago, between the beginning of Earth's accretion (~4.56 b.y.) and the age of the oldest Moon rocks brought back by the Apollo astronauts (4.47 b.y.). The latest period of heavy bombardment by meteorites ended about 3.9 b.y. ago.



We don't know exactly when life emerged on Earth, but chemical and geologic evidence suggest that primitive life may have been present by about 3.8 b.y., although scientists disagree about the exact timing. We do know that the Earth was a very different place at the dawn of life than it is today.

G2.11

Thought Question



Study the chart on p. 10 on the effects of impacts on Earth and life. Life on Earth must have emerged after the Moon-forming impact and before about 3.5 billion years ago, when there is good evidence for the existence of single-celled organisms. This observation implies that life evolved during, or just after, a very turbulent time in Earth's evolution.

Think about how impacts of different size might have influenced Earth during this time of heavy bombardment from about 4.5 to 3.9 b.y.

1. What would happen to the ocean in a “huge” impact compared with a “medium” impact?
2. What would happen to the atmosphere in a “huge” impact compared with a “medium” impact?
3. If life was present on Earth when a “huge” impact occurred, in what environment would it be mostly likely to survive?

12

[-< Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-10\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-10)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)

[Next -> \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-12\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-12)

G2.12



Summary

Be sure to review the main points of the lesson:

- Collision of a Mars-sized body with Earth about 4.51 b.y. ago
- Earth melted to its core, vaporizing or melting anything that existed
- Material was propelled from both bodies into space
- Impact sped up Earth's rotation and tilted Earth's axis by about 23° , which caused Earth to have seasons
- Earth reformed initially as a molten body, which then differentiated, with heavier elements sinking into the core and lighter elements coming to the surface, as it re-solidified
- Ejected material was caught in Earth's gravitational field, and aggregated and solidified to form the Moon

If you are unclear about any of the lesson content, review the eBook, videos and readings before class.

8

[← Back \(https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-11\)](https://catcourses.ucmerced.edu/courses/32116/pages/g2-dot-11)

[Geo1.2 TOC \(https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system\)](https://catcourses.ucmerced.edu/courses/32116/pages/geo1-origins-of-the-solar-system)